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RESEARCH ARTICLE

Multimodal Fake News Classification in Tamil Using Fact-Checked Social Media Content and Cost-Sensitive Learning

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ABSTRACT The exponential rise in user-generated social media content has contributed significantly to the spread of fake news. Regional language communities are particularly vulnerable because these fabricated narratives often integrate manipulated text, images, and audio, which we call multimodal fake news. Tamil language fake news detection presents unique challenges owing to the scarcity of multimodal datasets and the limited adaptation of state-of-the-art methods. This study proposes a transformer-based framework for multimodal fake news classification. It uses Bidirectional Encoder Representations from Transformers (BERT) because of its proven cross-lingual transfer capabilities and contextual understanding advantages in low-resource settings. We introduce TamilFacts, the first comprehensive multimodal dataset comprising 7,934 fact-checked samples, including text, images and speech, curated from Tamil fact-checking websites. The proposed approach integrates contextual embeddings for text, robust image feature extraction methods enhanced by error analysis, and denoised speech representations to capture multimodal context effectively. The experimental results demonstrate that transformer-based architectures efficiently classify fake news with an optimal feature representation and achieve an F1-score of 85%. Additionally, we integrated cost-sensitive learning to mitigate class imbalance by assigning higher weights to minority classes, thereby improving fairness and achieving an accuracy of 76.08%. This study establishes baseline benchmarks for Tamil multimodal fake news detection and provides methodological insights applicable to other regional Indian languages.

INDEX TERMS Fake news classification, low resource language, multimodal dataset, Tamil.

I. INTRODUCTION

Social media facilitates social connections and information exchanges based on shared interests and perspectives. However, its open and decentralized nature contributes to the rapid proliferation of misinformation [1], [2]. "Fake news" is a multifaceted concept that encompasses various forms of misinformation and disinformation. It is generally characterised by the intentional dissemination of false, misleading or fabricated information [3]. Creators strategically craft this misinformation, designing it to appear credible to manipulate public perception and propagate false narratives. [4]. Multimodal fake news, which integrates text, images, and videos, spreads rapidly, unlike text-based misinformation. Furthermore, people often perceive it as more trustworthy because this type of misinformation leverages the combined power of modalities to appear more realistic and evoke convincing emotions [5], [6]. The primary motivations behind misinformation include financial or political gain or intentional harm to individuals, organizations, or society [7].

Despite extensive research on misinformation in global contexts, which primarily focuses on the English language, studies on regional languages such as Tamil in the Indian context are limited [8]. Social media content such as memes

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in Tamil is prolific and highly effective medium for disseminating misinformation, making the language a critical area of study [9]. Addressing misinformation involves fact-checking, an effective process that verifies the accuracy of online news claims against facts and categorizes them as true or false [10]. In India, organizations dedicated to fact-checking encounter various hurdles, including difficulties in identifying and verifying local information owing to limited context-specific resources [11]. The scarcity of verified news content data has obstructed extensive research on low-resource languages, including Tamil [12].

To address these challenges, we propose TamilFacts as a comprehensive multimodal dataset for fake news classification in Tamil. This dataset consists of 7,934 fact-checked samples, which include text, image, and speech modalities curated from Tamil fact-check websites. Further, we have evaluated the performance of various transformer models to identify the most effective feature representation for multimodal fake news detection in low-resource language settings. By enabling systematic research on multimodal misinformation classification, this dataset serves as a benchmark resource for misinformation detection tasks in Tamil.

The proposed study used a comprehensive approach to evaluate the effectiveness of various models across different modalities. For the text modality, we utilized multiple transformer-based embedding models. Similarly, we employed a transformer model and various deep-learning models enhanced by error-level analysis for the image modality. For the speech modality, we explored traditional feature extraction techniques and transformer-based speech representations and incorporated denoising methods to improve the robustness. This experimental setup resulted in 96 unique combinations of embeddings, providing a comprehensive evaluation of multimodal feature representations.

We evaluated these combinations using multiple machine learning classifiers to identify the most effective multimodal fake news classification approach. Given the imbalanced nature of our dataset, we applied a cost-sensitive learning technique [13] to enhance classification performance. The evaluation metrics included the F1-score, accuracy, precision, and recall to assess model performance comprehensively.

In summary, the main contributions of this study are as follows:

- We present a novel multimodal dataset of fact-checked social media content for the Tamil language, consisting of 7,934 data samples spanning text, image, and speech modalities.
- We develop the first audio corpus in the Tamil language for the fake news domain based on fact-checked news articles, consisting of 884 minutes of audio, enabling multimodal analysis.
- We comprehensively evaluate multimodal transformer embedding models for misinformation classification in Tamil social media content.
- We introduce a cost-sensitive learning approach to mitigate class imbalance and reduce classification bias.
- We identify and evaluate a best performing multimodal embedding model for the developed dataset.

The rest of the paper is structured as follows: Section II reviews related work on multimodal fake news detection. Section III describes the curation process of the multimodal dataset, including data collection, dataset composition, class distribution, and the proposed methodology. Section IV presents the experimental details, while Section V concludes the paper with a summary of findings and future directions.

II. RELATED WORK

Research in multimodal fake news classification has advanced, driven by the proliferation of user-generated content on social media. In this section, we comprehensively review various approaches for Tamil fake news detection, considering low-resource Indian languages, resource-rich languages, and both unimodal and multimodal techniques.

Several studies have tackled fake news detection in Indian languages, mainly concentrating on textual data. For Urdu, Amjad et al. introduced UrduFND [14], a benchmark dataset of 900 articles and experiments using the AdaBoost model, with character-level and function word features, achieved a 0.87 F1-score, though the small dataset size and focus on lexical patterns limit generalizability. Phukan et al. pioneered Assamese fake news detection [15], evaluating LSTM (93.5% accuracy) and Bi-LSTM (94.2% accuracy) models on a 6,000-article dataset. Performance, while strong, was noted to be constrained by limited online Assamese resources.

In Hindi, Kumar and Singh [16] curated a 2,100-article dataset, achieving 92.36% accuracy with LSTMs. Their binary classification approach, however, overlooks partially misleading content. Badam et al. introduced Aletheia [17], a larger Hindi framework using 13,000 articles. Pre-trained Hindi BERT with Word2Vec features achieved 93.2% accuracy, surpassing TF-IDF, but the dataset's time-bound nature (2018-2021) limits event-driven performance evaluation. Garg et al. [18] addressed Hindi multimodal (text-image) fake news with an 8,000-sample dataset. While Bi-LSTM achieved 96.35% text accuracy and VGG-based image analysis contributed to 85.43% multimodal accuracy, the work omitted speech modality and lacked clarity on source credibility.

For Tamil, Mirnalinee et al. [19] contributed a 5,273-sample corpus, achieving 87.85% accuracy with an SVM baseline. Performance limitations stem from dataset imbalance and vocabulary overlap between classes. Devika et al. presented the first Malayalam dataset [20], categorized by severity. Logistic Regression on LaBSE embeddings yielded the best baseline F1-score (0.34), though the performance was heavily constrained by severe class imbalance and a small sample size (1,682 entries).

Beyond individual low-resource languages, significant efforts have focused on multimodal datasets and frameworks, often including English. Apoorva et al. [12] introduced the FakeNewsIndia dataset (4,803 incidents) and found YouTube engagement metrics predictive of fake videos (86-92% accuracy), unlike Twitter (37-41%). The study's English-only focus, however, limits generalizability within India. Singhal et al. presented FactDrill [21], a large-scale multilingual Indian dataset (22,435 samples, 13 languages) featuring an investigation_reasoning attribute. It enables cross-lingual analysis despite challenges like regional annotation variations and time-bound data (up to 2020). LekshmiAmmal and Madasamy [22] focused on Tamil multimodal (text-image) detection using approximately 3,500 samples from fact-checking sites. They employed Gemini 4 for image captioning and achieved a strong 0.87 F1-score with transformer models using ternary classification (TRUE, FALSE, FAKE). However, several gaps remain, such as modality coverage. The study excluded the audio modality, which is crucial for formats like podcasts and social media reels. Also, the dataset size (approx. 3,500 samples) is limited, potentially lacking diversity. The three-label classification does not differentiate between manipulated media (fabricated news) and misleading context (real news used falsely), an important difference for real-world application.

Several benchmark multimodal datasets have been proposed. Mishra et al. introduced FACTIFY [5] (50,000 image-text samples from India/US), showing multimodal analysis improved verification (Random Forest F1: 53.09%) but used basic models and binary labels. Suryavardan et al. extended this with FACTIFY 2 [23], adding satire and achieving 65% F1-score with BERT + Vision Transformers, though potential bias exists due to a strong political focus. Wang et al. introduced the LIAR dataset [24] (1,280 short statements), where a hybrid CNN using metadata slightly outperformed text-only models (0.274 accuracy), but the context was limited. Sharma and Garg [25] introduced IFND, an Indian text-image benchmark; LSTM-VGG16 fusion yielded 74% multimodal accuracy, but social media context was absent.

The multimodal feature fusion literature has advanced beyond simple concatenation toward techniques that capture complex nonlinear cross-modal dependencies. Attention-based approaches [26], including Transformer models with modality-specific tokens and contrastive learning, achieve stronger predictions by dynamically weighting features within and across modalities [27]. Complementing attention, modern fusion also includes tensor-based methods that model high-order cross-modal interactions, graph-based fusion that exploits relational structure via graph attention for dynamic, scalable fusion [28]. Recent surveys further argue for moving beyond early/intermediate/late fusion taxonomies toward principled, task-adaptive fusion strategies [29]. In parallel, audio manipulation and deepfake detection research emphasizes generalization: surveys highlight the need for robust, modality-aware detectors, and self-supervised encoders such as Wav2Vec 2.0 exhibit strong resilience to TTS-based manipulations [30]. For low-resource settings, cross-lingual transfer has proven effective, foundational work frames zero-shot transfer as emerging from cross-lingual alignment in multilingual models and analyzes how pretraining objectives induce language-neutral representations that support transfer [31]. Empirical evidence further shows that transfer is possible even without shared vocabulary when higher-layer parameters are shared, and that independently trained monolingual models can be aligned post hoc due to universal latent symmetries in embedding spaces [32]. In practice, pretrained multilingual models enable zero-shot transfer and markedly improve Dravidian-language NLP performance [33]. Finally, to address class imbalance common in multimodal datasets, cost-sensitive learning (CSL) preserves the original class distribution while minimizing high-cost errors by integrating misclassification costs directly into the loss [34]. Building on this foundation, Volk and Singer introduced adaptive CSL methods dynamically adjust the loss to correct train–test prior mismatches, improving reliability and efficiency under distribution shift [35].

Other approaches focus on platform-specific data or feature engineering. Mitra and Gilbert [36] developed Credbank for assessing Twitter event credibility using combined computational/human evaluation, though based on a limited sample. Valliappan and Ramya [37] demonstrated that integrating web-scraped metadata with LSTMs improved accuracy significantly up to 91%. Deepak and Chitturi [38] found BERT effective for fake review detection with an accuracy of 84.57%. Vinay et al. [39] achieved 90.3% accuracy on a Twitter dataset using Random Forest with rich features, noting social context improved detection by 18%. Amrita and Sankaran [40] used a trust-based framework for COVID-19 fake news, reaching 81.4% accuracy. Common limitations across these include potential dataset biases [38], labelling issues [39], and feature integration effectiveness [37].

Despite these efforts, studies have not yet explored multimodal fake news classification that integrates text, image, and speech modalities for low-resource Indian languages. Our work addresses a critical research gap by introducing the first trimodal (text, image, and speech) fake news classification system for a Dravidian language Tamil, moving beyond prior multimodal systems that have been limited primarily to unimodal or bimodal analysis. The following section details the creation of our comprehensive multimodal dataset, TamilFacts.

III. PROPOSED METHODOLOGY

This section presents the dataset collection pipeline and outlines the methodology employed for multimodal fake news classification. In addition, we elaborate on the details of the feature extraction methods, modality-specific models, machine learning classifiers, and the metrics used for evaluation.

A. DATASET

In this section, we outline the structure and key characteristics of our multimodal dataset, TamilFacts. Furthermore, we detail the methodology employed for collecting fact-checked news articles, the steps taken for preprocessing

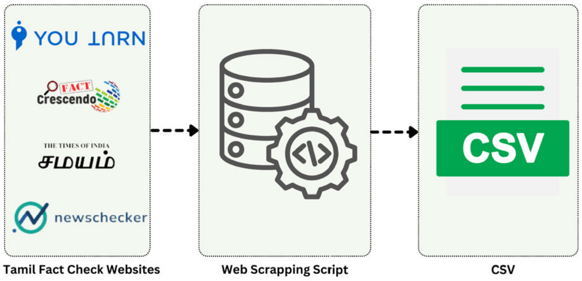

FIGURE 1. Data collection process from Tamil fact check websites.

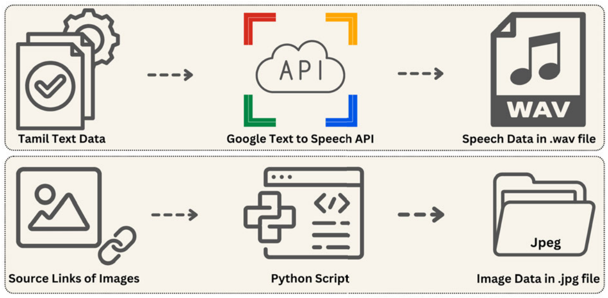

FIGURE 2. Curation of multimodal data.

the data, and the distribution of samples across different classes.

1) DATA COLLECTION

We developed our multimodal dataset by acquiring fact-checked news articles from four websites: youturn,[1] Tamil factscrescendo,[2] samayam Tamil,[3] and Tamil newschecker,[4] storing the data in a CSV file as shown in Fig. 1. These websites are well-known fact-checking organizations in Tamil Nadu. We deliberately selected our four fact-checking sources based on their established independence and non-partisan stance. This methodological step ensures the dataset remains free from significant political or regional biases. Some of these are International Fact-Checking Network[5] signatories, and some follow their guidelines. These websites publish fact-check news in Tamil, which aligns with our goal of collecting Tamil language data. They cover a wide range of topics, including politics, health, religion, history, sports, cinema, and recent events, thus ensuring a comprehensive collection. The websites provided structured, accessible content that could be scraped and used for research. We set a specific period for collecting articles from September 2017 to April 2024 to ensure we captured current and relevant content. We used filters to remove unverified content and ensured that we included only the validated information in the dataset. We identified fact-checked news articles by focusing on dedicated pages and categories within each website that contained fact-checked news content. These sections were typically labelled as "Fact Check", "Verified news", "To know more information", or similar terms in Tamil. Due to the unique HTML structure of each website, we used a custom web scraper script to acquire the data. Using Python's BeautifulSoup[6] library, data were automatically extracted from each news article and stored in a CSV file.

2) DATA COMPOSITION

Image data undergoes initial exploratory analysis. Images are downloaded from source links in the CSV file and stored in a dedicated folder as presented in Fig. 2, with filenames matching the corresponding text data IDs. A Python script validates and downloads each image using the source link, skipping corrupt images. This process resulted in a total of 7,934 downloaded images in JPG format. Simultaneously, the headline text from each fact-checked article in the CSV is converted into speech data using the Google Text-to-Speech API, configured to produce a natural Indian accent. The resulting Tamil audio data is saved in WAV format, yielding 7,934 audio samples. We chose Google TTS to generate a large-scale audio dataset for our experiments. However, we acknowledge that this process generates synthetic speech that does not capture the rich phonetic details, natural prosodic patterns (such as stress and rhythm), or diverse accents of authentic spoken Tamil. We therefore position this work as a foundational study within a controlled setting, accepting the use of synthetic data as a known constraint.

3) DATASET CLASS DISTRIBUTION

Based on the ratings provided by the authors of fact-checking articles across the four websites, the dataset is segmented into four distinct classifications, as detailed in Table 1: true, fake, altered and misleading. The fake category comprises the most significant portion of the dataset, with 5,841 samples (73.6%). Altered samples account for 441 (5.6%), misleading samples for 1,289 (16.2%), and true samples for 362 (4.6%). This distribution is dominated by fake news, providing insights into the spread of misinformation in Tamil. However, as illustrated in Fig. 3, This imbalance highlights the need for focused strategies to address and mitigate the prevalence of fake news effectively.

TABLE 1. Categories of class labels and their descriptions.

Labels	Description
True	The Claimed news is True according to the facts
Fake	The Claimed news is False according to the facts
Mis-leading	The news circulated contains both True and false information or Incomplete news that distorts facts to manipulate perception
Altered	Spread of edited or modified news as image cards or statements

Fig. 4 presents ideal examples from our multimodal dataset TamilFacts across four classification categories. The first

[1]https://youturn.in
[2]https://tamil.factcrescendo.com
[3]https://tamil.samayam.com
[4]https://newschecker.in/ta
[5]https://www.poynter.org/ifcn/
[6]https://pypi.org/project/beautifulsoup4/

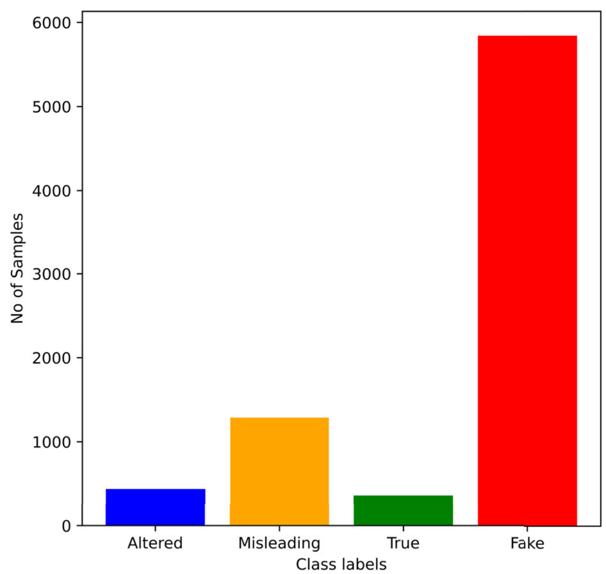

FIGURE 3. **Class distribution of the dataset and samples.**

column showcases a news sample classified as 'Fake', stating that "students in Chennai and Chengalpattu districts of Tamil Nadu are fighting after consuming drugs under the governance of the Tamil Nadu chief minister (referred to as 'Appa')." This news content and social media visuals show how fake news typically leverages imagery to support false narratives. The second column displays content classified as "True" featuring the Uttarakhand chief minister's statement that "cows are the only animals that exhale oxygen" This statement, given by him, is confirmed by the formal newspaper visible in the image. The third column presents a "Misleading" news sample containing elements of truth presented deceptively, showing the BJP Tamil Nadu leader Annamalai allegedly dancing to an item song during an election campaign, demonstrating that it isn't entirely false but contains misleading information to encourage incorrect conclusions. The final column illustrates the "Altered" category with a fabricated news card about Periyarist activists protesting at a salon. This card demonstrates how people (or one) can manipulate news contextually to create fabricated narratives. Each example includes the corresponding speech waveforms generated from the text, completing the multimodal representation that forms the basis of our feature extraction process.

4) DATA PREPROCESSING

We intentionally kept preprocessing minimal to preserve real-world authenticity. For the text modality, we used the original claim text provided in each fact-check article, the exact content circulating online, which fact-checkers have already verified. Consequently, no additional text cleaning was applied; this preserves linguistic features standard in online misinformation (e.g., code-mixing such as English words in Tamil script) so the model trains on realistic

Algorithm 1 An Algorithm Explaining the Workflow Used for Embedding Generation, Multimodal Fusion and Classification

Input: X = Text, Y = Image, Z = Speech.
Output: Trained machine learning models
Text models A = {mBERT, IndicBERT, LaBSE, MuRIL}
Image models B = {VGG16, ResNET50, ViT, VGG16+ELA, ResNET50+ELA, ViT+ELA}
Speech models C = {MFCC, HuBERT, MFCC+CAE, HuBERT+CAE}
Machine Learning Classifiers O={Random Forest, Logistic Regression, Decision Tree, Support Vector Machine, XGBoost, AdaBoost}
Initialize empty sets: $E_{\text{text}} \leftarrow \emptyset$, $E_{\text{image}} \leftarrow \emptyset$, $E_{\text{speech}} \leftarrow \emptyset$
for $x \in X$ **do**
  Process text sample x
  for $a \in A$ **do**
    $e_a(x) \leftarrow \text{TextModel}_a(x)$
    $E_{\text{text}} \leftarrow E_{\text{text}} \cup \{e_a(x)\}$
  end for
end for
for $y \in Y$ **do**
  Process image sample y
  for $b \in B$ **do**
    $e_b(y) \leftarrow \text{ImageModel}_b(y)$
    $E_{\text{image}} \leftarrow E_{\text{image}} \cup \{e_b(y)\}$
  end for
end for
for $z \in Z$ **do**
  Process speech sample z
  for $c \in C$ **do**
    $e_c(z) \leftarrow \text{SpeechModel}_c(z)$
    $E_{\text{speech}} \leftarrow E_{\text{speech}} \cup \{e_c(z)\}$
  end for
end for
Let $E_{\text{fused}} \leftarrow \emptyset$ {Initialize fused embeddings collection}
for $(e_t, e_i, e_s) \in E_{\text{text}} \times E_{\text{image}} \times E_{\text{speech}}$ **do**
  FusedEmbedding $\leftarrow [e_t \| e_i \| e_s]$ {Concatenation}
  $E_{\text{fused}} \leftarrow E_{\text{fused}} \cup \{\text{FusedEmbedding}\}$
end for
for each fused_embedding **e** in E_{fused} **do**
  for each classifier *model* in $\mathcal{O}$ **do**
    Train the model using:

$$\underbrace{model_{\text{trained}}}_{\text{Final classifier}} = \text{Train}\left(\underbrace{\mathbf{e}}_{\text{Features}}, \underbrace{y}_{\text{Labels}}\right)$$

    Save $model_{\text{trained}}$ for future use
  end for
end for

data. We used the exact verified claim text to generate synthetic speech via Google Text-to-Speech (TTS). For the image modality, we applied only essential preprocessing for model compatibility, resizing to the required input dimensions while preserving crucial visual information such as potential manipulation features, prior to feature extraction.

B. MULTIMODAL FEATURE EXTRACTION WITH TRANSFORMERS AND CONVOLUTIONAL NETWORKS

Modern fake news detection demands robust handling of inconsistencies across text, image, and speech modalities.

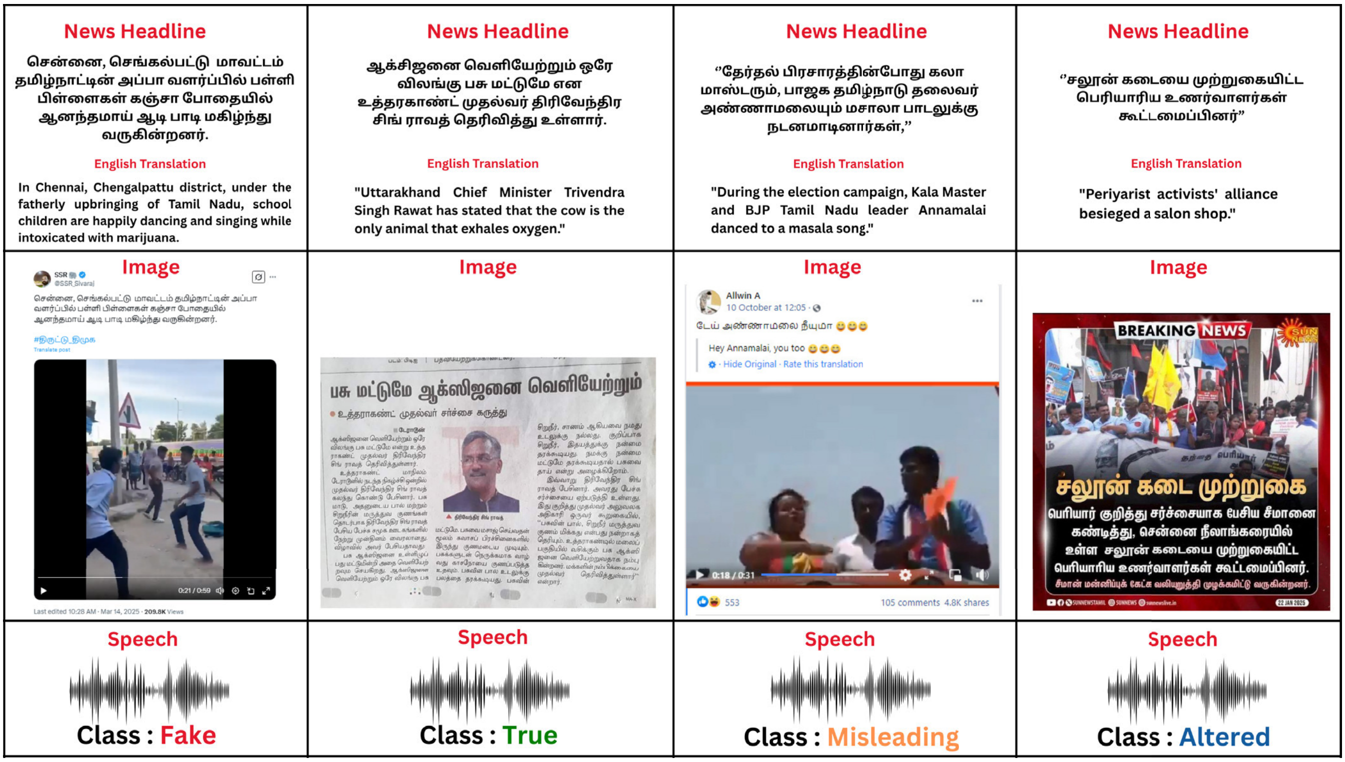

FIGURE 4. Dataset samples from the dataset.

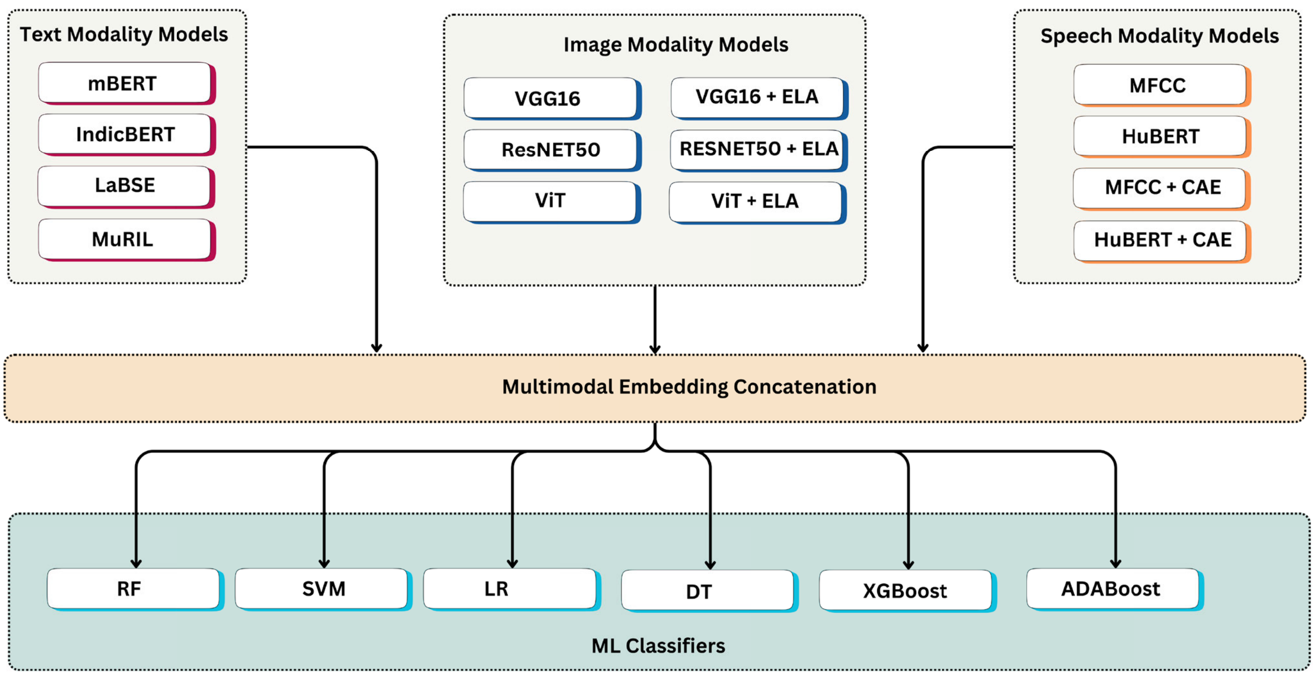

FIGURE 5. Proposed architecture for the classification of multimodal fake news in Tamil language.

This requirement necessitates models that can differentiate subtle contextual relationships, which unimodal approaches often miss. The traditional recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are capable of retaining prior information, struggle with long-range dependencies and are limited by their sequential processing nature, which hinders parallel computation [41]. To overcome these limitations, we employ a methodology that leverages transformer-based architectures and Convolutional Neural Networks (CNNs) to enable efficient feature extraction across modalities, capturing local and global relationships crucial for identifying manipulated content.

As illustrated in Fig. 5, the proposed approach integrates feature extraction across textual, visual, and audio modalities, multimodal fusion, and classification. This architecture leverages state-of-the-art models (SOTA), including transformer-based language models (mBERT [42], IndicBERT [43], LaBSE [44], MuRIL [45]), Convolutional Neural Networks (VGG16 [46], ResNet50 [47]) and Vision Transformers [48] for image analysis, and Hidden-Unit BERT (HuBERT) [49] along with the traditional signal processing Mel-Frequency Cepstral Coefficients (MFCC) [50] technique for speech analysis.

The Transformer model, leveraging self-attention mechanisms, offers a more effective architecture for managing long-range dependencies and facilitating parallelization [26]. This BERT model consists of encoder-decoder stacks that allow the model to simultaneously attend to multiple locations in the input sequence, thereby capturing a wide array of contextual relationships. Integrating positional encodings preserves the sequential structure of the input data, ensuring the model understands the order of tokens.

C. MODALITY-SPECIFIC FEATURE EXTRACTION

This section details the feature extraction methods used for each modality. Feature extraction identifies and selects the most important characteristics from raw data, such as text, images, or audio. This step transforms the complex raw data into a simpler format that machine learning models can use for classification.

Text Modality: We employ BERT-based models, including mbert [42], Indicbert [43], Labse [44], and Muril [45], to generate contextualised word embeddings. BERT-based models are particularly effective for detecting deceptive language patterns in news headlines and articles because they capture semantic nuances and cross-lingual relationships [51].

Image Modality: We use three models (VGG16 [46], ResNet50 [47], and Vision Transformer [48]) to extract image features. Convolutional models such as VGG16 and ResNET50 identify digital tampering of images with ELA, highlighting inconsistencies in image quality [52]. The ViT analyzes the entire image to capture global patterns and relationships, complementing the localized features from convolutional models.

Speech Modality: We use two main techniques to analyze the audio data. First, we extract Mel-Frequency Cepstral Coefficients (MFCCs) [50] to represent the basic sound features of speech. MFCCs are effective for identifying speakers and analyzing emotions in speech [53]. Second, we use a more advanced model called Hidden Unit BERT (HuBERT) [49] to capture more complex patterns in the speech. HuBERT is better at understanding the context and subtle changes in a speech that might indicate deception [54]. We use a convolutional denoising autoencoder (CDAE) [55] to make the analysis more robust. This tool helps to remove unwanted noise from the speech data, making the important features clearer. By combining these three methods, MFCCs, HuBERT and CDAE, we were able to get a comprehensive view of the speech data. This approach lets us detect evident and subtle signs of manipulation or deception in the audio content.

D. TEXT MODALITY

This subsection explains the different models we used to analyze the text data and extract its key features.

1) mBERT

We used multilingual BERT (mBERT) [42], specifically the bert-base-multilingual-cased variant, chosen for its robust cross-lingual capabilities essential for processing Tamil text in our dataset which contains Tamil news articles. Pre-trained on corpora from 104 languages, mBERT effectively leverages knowledge learned from high-resource languages to enhance performance in languages with less available data, such as Tamil. This transfer learning capability is particularly valuable given the relative scarcity of large, annotated Tamil datasets for specific tasks like fake news detection. The model employs a standard Transformer architecture consisting of 12 layers, 768 hidden units per layer, and 12 self-attention heads. This architecture facilitates the model to capture complex contextual relationships between words within the input text. For input processing, mBERT uses the WordPiece tokenization algorithm. WordPiece segments text into subword units based on a vocabulary derived from the pre-training corpus. This subword approach is advantageous as it can handle words not seen during training (out-of-vocabulary words) by breaking them into known pieces. It is beneficial for morphologically rich languages and helps manage the diverse or informal vocabulary often found in Tamil social media content. WordPiece operates by greedily finding the longest possible subword prefix from its vocabulary that matches the beginning of the remaining word segment. To obtain a single, fixed-dimension vector representing the entire sentence, we applied mean pooling to the final hidden states of all tokens generated by mBERT for that sentence. This involves calculating the element-wise average of the token vectors.

To obtain a single, fixed-dimension vector representing the entire sentence, denoted as $\mathbf{e}_{\text{sent}}$, we applied mean pooling to the final hidden states of all tokens generated by mBERT. We compute this embedding as follows:

$$\mathbf{e}_{\text{sent}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{h}_i \tag{1}$$

where $\mathbf{h}_i$ represents the final hidden state vector for the i-th token, and n is the total number of tokens in the sequence (sequence length). This mean-pooled embedding consolidates contextual information from all tokens, providing a complete semantic representation of the sentence.

2) IndicBERT

We incorporated IndicBERT [43], a pre-trained model based on the ALBERT architecture [56], specifically optimized for computational efficiency and performance across various Indian languages, including Tamil. IndicBERT leverages architectural optimizations inherent to ALBERT, such as cross-layer parameter sharing. This method decreases the number of model parameters compared to traditional BERT models. It makes computing faster without losing the model's ability to understand data. The specific configuration employed utilizes a hidden size of 768, 12 hidden layers, and 12 attention heads. The creators pre-trained the model on a multilingual corpus with 12.7GB of Tamil text sourced from IndicCorp [43] which is highly relevant for our task. The pre-training relied on the masked language modelling (MLM) objective, where the model learns contextual representations by predicting randomly masked tokens within input sequences. We extracted the embedding associated with the special `[CLS]` token for downstream tasks requiring a sentence-level representation, such as classifying news headlines. We prepend this token to every input sequence during processing. We use its corresponding final hidden state vector, $\mathbf{h}_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$, in BERT-based models, as the pre-training process optimizes it to aggregate sequence-level information, providing a suitable input vector for a classification layer.

3) MuRIL

We also used MuRIL (Multilingual Representations for Indian Languages) [45], a BERT-based model engineered explicitly for Indian languages, demonstrating superior performance compared to mBERT on several Indic NLP benchmarks. MuRIL's effectiveness stems from its unique pre-training approach. Its creators concurrently trained it on monolingual corpora for 17 Indian languages (including Tamil) and leveraged transliterated text pairs sourced from datasets like PMINDIA [57] and Dakshina [58]. This inclusion of transliterated data enables the model to learn robust cross-script mappings and relationships, which is particularly advantageous for our context, as users often employ multiple scripts for the same language. For Tamil, this results in an enhanced ability to capture context even when encountering romanized or mixed-script text. Similar to standard BERT-based approaches for classification, MuRIL utilizes the final hidden state vector corresponding to the special `[CLS]` token ($\mathbf{h}_{[\mathrm{CLS}]}$) as a primary representation for the entire input sequence. The description provided suggests a subsequent refinement step involving positional encodings ($\mathbf{p}$) and Layer Normalization, formulated as:

$$\mathbf{e}_{\mathrm{muRIL}} = \mathrm{LayerNorm}(\mathbf{h}_{[\mathrm{CLS}]} + \mathbf{W}_p\mathbf{p}) \tag{2}$$

Here, $\mathbf{h}_{[\mathrm{CLS}]}$ is the final hidden state of the `[CLS]` token, $\mathbf{p}$ represents positional encoding vectors, $\mathbf{W}_p$ likely denotes a learned projection matrix for these encodings, and `LayerNorm` refers to the Layer Normalization operation. This equation suggests combining the sequence representation with weighted positional information before applying normalization to yield the final MuRIL embedding ($\mathbf{e}_{\mathrm{muRIL}}$). The enhanced contextual comprehension offered by MuRIL assists in detecting semantic inconsistencies, which may suggest fabricated content, such as possibly altered quotes in Tamil political news articles.

4) LaBSE

We utilized the Language-agnostic BERT Sentence Embedding (LaBSE) model [44] to generate sentence embeddings designed explicitly for high performance on cross-lingual tasks. LaBSE aims to map sentences from different languages into a shared, high-dimensional semantic space.

LaBSE employs a dual-encoder Transformer architecture. This architecture involves two parallel BERT-based encoders that the creators trained jointly using translation ranking tasks. One of the encoders processes input text, in our context Tamil news articles headline, to produce a fixed-dimension embedding vector (typically 768-dimensional). The training objective encourages sentences with similar meanings (like translation pairs) across different languages to have nearby representations in the shared embedding space.

A key strength of LaBSE is its ability to assess semantic similarity between sentences irrespective of their source language. We can quantify this by computing the cosine similarity between the embedding vectors of two sentences.

If $\mathbf{e}_1$ and $\mathbf{e}_2$ are the LaBSE embedding vectors for two sentences, we calculate their similarity as:

$$\mathrm{Similarity}(\mathbf{e}_1, \mathbf{e}_2) = \frac{\mathbf{e}_1 \cdot \mathbf{e}_2}{\|\mathbf{e}_1\|\|\mathbf{e}_2\|} = \cos(\theta) \tag{3}$$

where $\cdot$ denotes the dot product, $\|\cdot\|$ represents the Euclidean norm (vector magnitude), and θ is the angle between the two vectors in the embedding space. The model's capability to develop cross-lingual embeddings helps to identify claims that may be misleading due to mistranslations.

E. IMAGE MODALITY

This section briefs about the models and techniques used for extracting features of image modality.

1) VISION TRANSFORMER

We utilize Vision Transformer (ViT) to capture long-range dependencies in image patches. Splitting images into fixed-size patches (16 × 16 or 32 × 32 pixels), ViT applies transformer-based attention to identify subtle manipulation artifacts often missed by CNNs. Specifically, we compute visual anomaly scores using the attention weights:

$$\text{Anomaly Score} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}|A_{ij} - \overline{A}| \quad (4)$$

where A_{ij} is the attention weight between patch i and j, $\overline{A}$ is the mean attention weight, and N is the number of patches. High anomaly scores indicate irregular patch relationships, potentially signifying tampering in fabricated images. The $\mathbf{h}_{[CLS]}$ token is used to obtain a global image representation $\mathbf{e}_{\text{ViT}}$ for downstream fusion.

2) VGG16

We employ VGG16 [46] to extract foundational visual features. Its deep architecture, characterized by 13 convolutional layers using small 3×3 kernels arranged in blocks, followed by max-pooling layers for spatial downsampling, enables the hierarchical extraction of features. Early layers detect simple patterns like edges and textures, while deeper layers capture more complex object shapes and parts. Such visual features can indicate potential manipulation when semantically incongruent with accompanying text (e.g., a generic stock photo associated with a specific breaking news headline). Given an input image I, we extract the VGG16 embedding $\mathbf{e}_{\text{VGG16}}$ from the output of the final max-pooling layer (typically after the fifth convolutional block) before the fully connected layers:

$$\mathbf{e}_{\text{VGG16}} = \text{GlobalAvgPool}(\text{Conv}_{\text{block5\_pool}}(I)) \quad (5)$$

where $\text{Conv}_{\text{block5\_pool}}(I)$ denotes the feature map output from the last max-pooling layer within the convolutional base. This resulting feature vector, $\mathbf{e}_{\text{VGG16}}$, encapsulates high-level visual semantics derived from the input image, suitable for downstream analysis.

3) ResNet50

ResNet50 addresses the vanishing gradients problem via residual connections, enabling deeper analysis of image manipulations. We leverage this architecture, pre-trained on ImageNet, to identify complex deepfake artifacts in Tamil entertainment endorsements and political images. Given an input image I, the ResNet50 embedding is:

$$\mathbf{e}_{\text{ResNet50}} = \text{AvgPool}(\text{ResBlock}_{50}(I)) \quad (6)$$

We calculate the feature-level similarity between the input image and a known authentic image of the subject using cosine similarity:

$$\text{Similarity} = \cos(\mathbf{e}_{\text{ResNet50}}, \mathbf{e}_{\text{authentic}}) \quad (7)$$

Low similarity scores suggest potential deepfake manipulations.

F. ERROR LEVEL ANALYSIS

We used error level analysis (ELA) to detect subtle image tampering by identifying inconsistencies in compression distortion. ELA operates by re-saving an image at a specific quality level and comparing it to the original, highlighting regions with divergent compression artifacts indicative of editing, additions, or alterations. This technique is beneficial for detecting manipulated images circulating within social media and news outlets, where sophisticated editing tools might be used to spread misinformation [52].

ELA enhances forensic scrutiny of image integrity by exposing variations in compression levels across image regions, often imperceptible to the naked eye. In our framework, ELA is a preprocessing step before feature extraction using ViT, VGG16, and ResNet50, allowing us to focus on regions identified as potentially tampered.

To quantify the impact of ELA, we compute a difference map, D, between the original image I and its ELA-processed version I_{ELA}:

$$D = |I - I_{ELA}| \quad (8)$$

This difference map is then used as a mask to weigh the feature embeddings generated by ViT, VGG16, and ResNet50:

$$\mathbf{e}_{modified} = \mathbf{e}_{original} \odot (1 + \lambda D) \quad (9)$$

where:

- $\mathbf{e}_{original}$ is the original embedding from ViT, VGG16, or ResNet50.
- $\odot$ represents element-wise multiplication.
- λ is a scaling factor that controls the influence of the ELA difference map.

Integrating the Error Level Analysis (ELA) difference map into feature embeddings enhances the model's sensitivity to tampered regions by emphasizing features associated with potential manipulations. The difference map (D) captures compression inconsistencies between the original image and its recompressed version, highlighting areas with differing compression levels, which may indicate tampering. Larger values in D suggest a higher likelihood of manipulation. Our approach uses this map (D) to weight the feature embeddings, which directs the model's focus toward suspicious regions. The scaling factor (λ) determines the degree of emphasis, ensuring that our method prioritizes regions flagged by ELA in the modified embedding ($\mathbf{e}_{\text{modified}}$). By amplifying these ELA-derived signals, we better equip the model to identify subtle manipulations within images.

G. SPEECH MODALITY

This section briefs about the models and techniques used for extracting features of speech modality.

1) HuBERT

We utilize HuBERT (Hidden-unit BERT) [49] to detect subtle anomalies in synthetic speech, potentially generated from misleading Tamil headlines. By predicting masked speech segments during pre-training, HuBERT learns robust acoustic representations that effectively capture deviations from natural human speech patterns [54]. The ability to capture features of system-generated audio is important. Synthetic speech showcases variations in rhythm, tone, and pronunciation accuracy.

Given a sequence of input speech frames, S, HuBERT generates contextualized hidden unit representations, $H \in \mathbb{R}^{T \times d}$, where T is the sequence length and d is the hidden dimension (e.g., 768 for HuBERT-Base). To derive a single utterance-level embedding, we apply mean pooling across the time dimension:

$$\mathbf{e}_{\text{HuBERT}} = \frac{1}{T} \sum_{t=1}^{T} H_t \tag{10}$$

where H_t is the hidden representation for frame t. This resulting embedding, $\mathbf{e}_{\text{HuBERT}}$, encapsulates the overall acoustic characteristics of the speech utterance, including potential indicators of synthesis.

2) MFCC

We extract Mel-Frequency Cepstral Coefficients (MFCCs) to represent the spectral envelope of the synthetic speech signals. MFCCs mimic human auditory processing, capturing short-term power spectra through a series of transformations: pre-emphasis, framing, windowing, Fast Fourier Transform, Mel-scale filtering, logarithmic compression, and Discrete Cosine Transform. We concatenate the MFCCs across time to create a feature vector:

$$\mathbf{e}_{\text{MFCC}} = [\text{MFCC}_1, \text{MFCC}_2, \ldots, \text{MFCC}_F] \tag{11}$$

where F is the number of frames. We then compute statistical features such as mean, variance, and range from these MFCCs for each utterance to capture the overall spectral characteristics. We comprehensively understand the characteristics of synthetic speech by analyzing the MFCCs in conjunction with HuBERT embeddings.

H. CONVOLUTIONAL AUTOENCODER FOR DENOISING SPEECH DATA

We employ a convolutional autoencoder (CAE) to reduce noise and artifacts in synthetic speech data generated from Tamil text headlines. Using a CAE proves crucial because even advanced TTS systems produce synthetic speech with subtle acoustic irregularities that can confound feature extraction. Furthermore, real-world audio often includes background noise. By denoising the audio before extracting features with HuBERT and MFCC, we aim to improve our fake news detection system's robustness.

The CAE contains two main components: an encoder that compresses noisy input into a lower-dimensional latent representation and a decoder that reconstructs a denoised version. The encoder uses convolutional layers to extract hierarchical spatial features, while the decoder uses deconvolutional layers to reconstruct essential features while suppressing noise.

We train the model on pairs of noisy and clean speech samples to minimize reconstruction error. The training process mathematically operates as follows:

Let X represent a clean speech signal and X' its noisy counterpart. The encoder E maps X' to latent representation Z:

$$Z = E(X') \tag{12}$$

The decoder D then reconstructs the signal:

$$\hat{X} = D(Z) = D(E(X')) \tag{13}$$

We optimize the CAE to minimize mean squared error (MSE) between original X and reconstructed $\hat{X}$:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^{N} (X_i - \hat{X}_i)^2 \tag{14}$$

where N is the training set's sample count.

After training, we use the CAE to denoise speech signals before feature extraction. This denoised input enables HuBERT and MFCC models to produce more accurate feature embeddings.

I. MULTIMODAL EMBEDDING CONCATENATION

We constructed 96 unique multimodal embeddings by systematically concatenating features across text, image, and speech modalities Fig. 6. This cross-modal fusion combines four text embeddings, six image embeddings, and four speech embeddings through direct vector concatenation, preserving the distinct information patterns from each modality while enabling joint representation learning.

Text modality embeddings derive from four multilingual BERT variants (mBERT, IndicBERT, LaBSE, MuRIL), image embeddings from six vision models (VGG16, Vision Transformer, ResNet50, each with/without ELA), and speech embeddings from four acoustic representations (MFCC, HuBERT, and their denoised counterparts). The Cartesian product of these modality-specific embeddings ($4 \times 6 \times 4$) yields 96 combinatorial variants, allowing comprehensive analysis of cross-modal interactions for fake news classification.

J. MACHINE LEARNING CLASSIFIERS

We employed six machine learning algorithms: Random Forest [59], Support Vector Machine [60], Decision Tree [61], Logistic Regression [62], XGBoost [63], and AdaBoost [64] to evaluate 96 multimodal embedding combinations for multimodal fake news classification. We selected these models for their demonstrated effectiveness in handling multimodal data distributions.

1) Random Forest employs ensemble decision trees that generate predictions through majority voting. The algorithm utilizes bootstrap aggregating (bagging) to train individual trees on random data subsets while randomly selecting a predefined number of features at each node split, enhancing model robustness and reducing overfitting.
2) The Support Vector Machine (SVM) identifies an optimal hyperplane to maximize the margin between

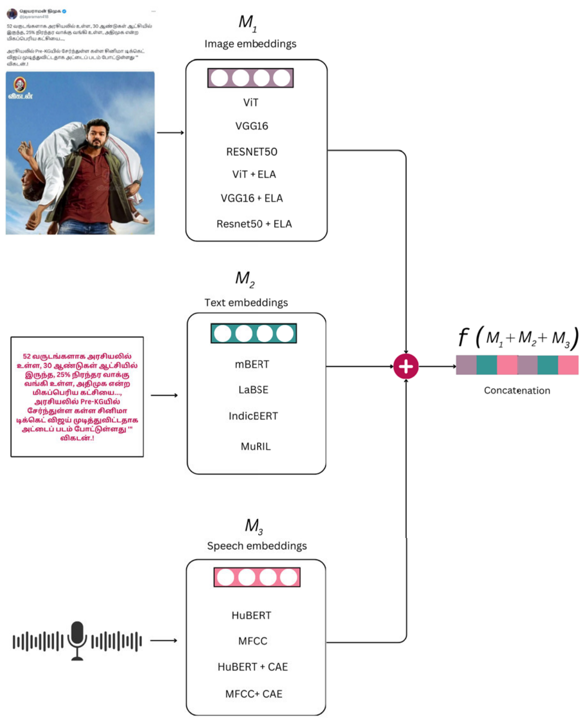

FIGURE 6. Concatenation process of text, image and speech data.

distinct classes in a high-dimensional feature space. This algorithm performs binary classification and addresses non-linear separability through kernel functions, efficiently handling datasets of up to 1,000 samples [65].

3) A decision tree uses a supervised learning method that recursively partitions data into subsets using feature-value thresholds. The algorithm determines splits through optimization criteria Gini impurity or information gain, which measures class separability. This method constructs a hierarchical structure where internal nodes represent decision rules and terminal leaf nodes assign final class labels. The transparent model architecture allows an intuitive interpretation of classification pathways.
4) Logistic regression provides a statistically robust framework for binary classification by modelling class probabilities through a sigmoid transformation of linear input combinations. The algorithm estimates sample membership likelihoods using a sigmoid function that maps log odds to probability intervals. The algorithm optimizes model parameters through maximum likelihood estimation, iteratively adjusting coefficients to maximize the probability of observing the training data under the learned distribution.
5) XGBoost (Extreme Gradient Boosting) employs a gradient-boosted ensemble technique that builds decision trees sequentially in an additive manner. The algorithm's parallelized architecture and histogram-based splitting further ensure computational efficiency, making it particularly suited for high-dimensional multimodal data and scalable to large-scale datasets.
6) AdaBoost (Adaptive Boosting) iteratively combines multiple weak learners, typically decision stumps, through an ensemble algorithm. During training, it dynamically increases the weights of misclassified samples in successive iterations, forcing subsequent learners to focus on more complex cases. The algorithm determines the final prediction through a weighted majority vote of all weak learners, assigning higher weights to more accurate base models.

K. EVALUATION METRICS

We evaluate classifier performance using four metrics: accuracy, precision, recall, and F1-score, derived from the confusion matrix. This matrix categorizes predictions into four classes: true positives (TP: correctly identified positive cases), true negatives (TN: correctly rejected negative cases), false positives (FP: negative cases misclassified as positive), and false negatives (FN: positive cases misclassified as negative).

Accuracy measures overall prediction correctness (TP+TN divided by total predictions).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

Precision quantifies prediction reliability (TP proportion among all positive predictions).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

Recall (sensitivity) assesses detection completeness (TP proportion among actual positives).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

The F1-score balances these metrics through their harmonic mean, addressing scenarios where precision-recall tradeoffs require optimization, particularly crucial for imbalanced fake news detection tasks.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

IV. EXPERIMENTS AND RESULTS

This section presents the results of our multimodal fake news classification experiments. We evaluate six classifiers across 96 multimodal embedding combinations using accuracy, precision, recall, and F1-score as key performance metrics. To ensure uniformity, we report all metrics in percentage-scaled form. Table 2 summarizes the findings from these experiments, providing a comprehensive comparison of the top four multimodal embedding combinations. While Table 2 highlights optimal configurations, we provide the complete set of 96 experiments in Table 7 of Appendix. Furthermore, to validate the impact of our multimodal approach, we compare its performance against a unimodal baseline. We use a paired t-test to determine the statistical significance of any performance gains.

TABLE 2. Classification results of top four performing multimodal embedding combinations with 95% confidence intervals.

Text Embedding	Speech Embedding	Image Embedding	Classifier	Accuracy %	F1 Score %	Precision %	Recall %
IndicBERT	HuBERT	ViT + ELA	DT	75.80 [74.12, 77.48]	85.00 [82.00, 88.00]	90.00 [87.00, 93.00]	95.00 [93.00, 97.00]
IndicBERT	HuBERT	ResNet + ELA	LR	76.08 [74.42, 77.74]	85.00 [82.00, 88.00]	85.00 [82.00, 88.00]	84.00 [81.00, 87.00]
IndicBERT	MFCC	ViT	LR	74.95 [73.25, 76.65]	85.00 [82.00, 88.00]	86.00 [83.00, 89.00]	87.00 [84.00, 90.00]
IndicBERT	Denoised_MFCC	VGG16	DT	75.66 [73.98, 77.34]	85.00 [82.00, 88.00]	92.00 [89.00, 95.00]	79.00 [76.00, 82.00]

A. EXPERIMENTAL SETUP

We split our dataset of 7934 multimodal samples into an 80% training set (6347 samples) and a 20% test set (1587 samples). To ensure a fair evaluation, we used stratified sampling to maintain the same class proportions in both sets. We also set a fixed random seed (random_state=42) to ensure our results are fully reproducible. Our experiments were conducted on an AI cloud platform from Jarvis Labs, using an NVIDIA RTX 5000 GPU, 32 GB of RAM, and 7 CPU cores. For our implementation we used several key open-source libraries, including PyTorch,[7] Sentence-Transformers,[8] and scikit-learn.[9] This work represents the first comprehensive study on fake news detection in the Tamil language to integrate all three modalities such as text, image, and speech. Consequently, direct, comparable baselines are not available in the existing literature, and we therefore position our state-of-the-art, transformer-based approach as a foundational benchmark to facilitate future research in this important and low-resource domain.

B. COST SENSITIVE LEARNING

As illustrated in Figure 3 shows the class distribution, highlighting the dominance of the Fake class (73.6%) relative to Misleading (16.2%), Altered (5.6%), and True (4.6%). Our dataset exhibits significant class imbalance, with the "Fake" category disproportionately dominating the distribution. This imbalance causes classifiers to prioritize majority-class predictions while neglecting minority classes. To counter this bias, we implement cost sensitive learning (inverse frequency weighting) to equalize class influence during training. We calculate class weights as:

$$w_i = \frac{n}{k \cdot n_i} \tag{19}$$

Here, w_i denotes the weight for class i, n represents the total number of samples, k is the number of classes, and n_i is the number of samples in class i. This strategy assigns higher weights to underrepresented classes such as "True," "Misleading," and "Altered," directly adjusting the loss function to impose higher penalties for minority-class misclassifications. By prioritizing balanced error correction across all categories, the model learns to recognize minority patterns more effectively without overfitting to dominant classes.

Fig. 7 and Fig. 8 compare confusion matrices before and after applying cost-sensitive learning, revealing substantial

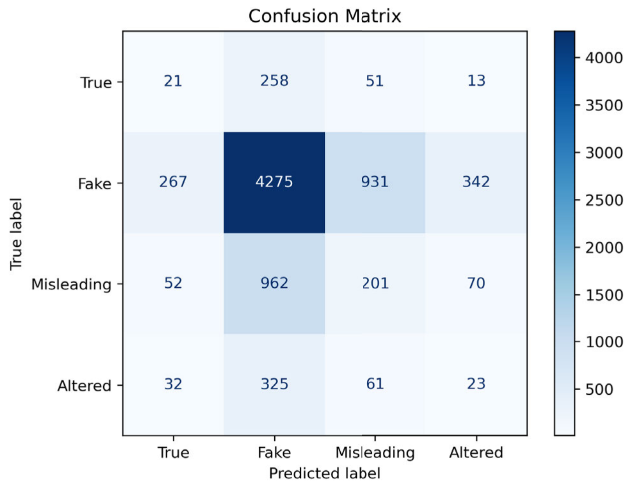

FIGURE 7. Classification performance of the best performing embedding "IndicBERT+HuBERT+ViT_ELA" without incorporating cost sensitive learning.

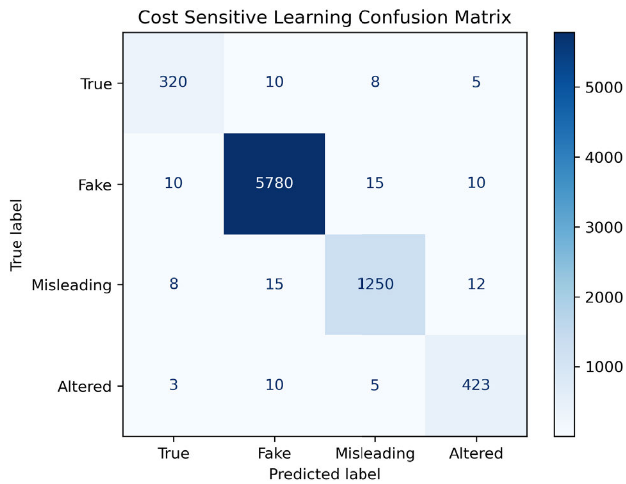

FIGURE 8. Classification performance of the best performing embedding "IndicBERT+HuBERT+ViT_ELA" after incorporating cost sensitive learning.

performance improvements. Prior to implementation Fig. 7, the model exhibited strong bias toward the "Fake" class, correctly classifying 4,275 instances but misclassifying 258 "True" and 962 "Misleading" instances as "Fake." This reflects insufficient feature discrimination and class imbalance dominance. After cost-sensitive adjustments, Fig. 8, the overall F1 Score improved by 15%, with notable gains in minority classes:

1) "True" class: Correct identifications surged from 21 to 320 instances.

[7] PyTorch version 2.4, https://pytorch.org/
[8] Sentence-Transformers version 3.0, https://www.sbert.net/
[9] scikit-learn version 1.6, https://scikit-learn.org/

2) "Misleading" class: Correct classifications increased from 201 to 1,250 instances.
3) "Altered" class: Accurate predictions increased from 23 to 423 instances.
4) "Fake" class: While predictions grew from 4,275 to 5,780, this reflects reduced bias, as misclassifications of non-"Fake" instances dropped sharply.

By applying cost-weighting, we dramatically boosted performance in the "Altered" class, increasing correct predictions from 23 to 423. This change made the model more aware of minute signs of visual tampering, which our selected image models captured but the system had previously ignored. We also saw improvement in the "Misleading" class, as the model learned to spot mismatches between features of text, image and speech. These findings show that useful features for detecting minority class types of misinformation (altered, misleading) already existed in our data embeddings. The cost-sensitive learning approach forced the model to pay attention to them by penalizing weights. At the same time, this led to a slight drop in accuracy for the "Fake" class. However, the model now shifts from a biased, high-precision model to a broader, unbiased model that captures all types of misinformation.

These results demonstrate that cost-sensitive learning effectively mitigates class imbalance, reduces majority-class overfitting, and enhances the model's capacity to differentiate subtle patterns in underrepresented categories. The technique achieves a more equitable accuracy distribution (90% across classes) while maintaining robust performance in the majority class.

C. BEST PERFORMING MULTIMODAL EMBEDDINGS

This section shows how well our fake news detection model works. After running 96 tests with multiple combinations, we found the four best models for a thorough analysis. To better understand each model, we studied three key things: (1) how well it learned using overfitting curves, (2) how accurately it classified news using ROC curves, and (3) how it groups information using 3D t-SNE visualizations.

In this section, we employ three key visualizations to comprehensively evaluate our model's performance. First, overfitting curves illustrate how the model learns across 20 training epochs, with the F1 score as the evaluation metric due to class imbalance in the dataset. The F1 score combines precision and recall into a single value, making it more useful than accuracy when the data is imbalanced. In addition, ROC curves show how well a classifier works by plotting the True Positive Rate (how many real positives the classifier correctly identifies) against the False Positive Rate (how many negatives the classifier incorrectly labels as positive) at different threshold settings. This graph helps us see the balance between detecting true cases and avoiding false alarms, making it easier to choose the best cutoff point for classification. Finally, t-SNE 3D plots provide insights into how the model organizes high-dimensional data into clusters based on text, image, and speech components.

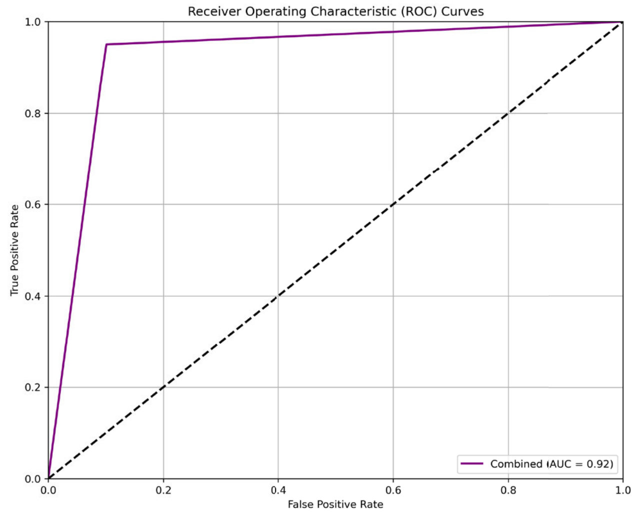

FIGURE 9. ROC curve of IndicBERT + HuBERT + ViT_ELA embedding.

This analysis gave us a complete picture of the model's learning, its ability to tell real news from fake, and how it uses information. Below, we show the results for the four best models: "IndicBERT + HuBERT + ViT_ELA", "IndicBERT + HuBERT + ResNet_ELA", "IndicBERT + MFCC + ViT", and "IndicBERT + Denoised_MFCC + VGG16" showing how well the model uses different types of information to detect fake news.

1) IndicBERT + HuBERT + ViT_ELA

The "IndicBERT + HuBERT + ViT_ELA" combination demonstrates exceptional performance, achieving an Area Under the Curve (AUC) score of 0.92, as illustrated by its ROC curve (Figure 9). This high AUC value indicates that the model effectively distinguishes between classes, with a strong balance between sensitivity (TP) and specificity (FP). The curve closely approaches the top-left corner of the plot, signifying minimal trade-offs between false positives and false negatives. This result highlights the robustness of this embedding combination in capturing meaningful multimodal features for fake news detection.

As shown by the overfitting curve in Figure 10, training accuracy steadily increases across epochs, reaching near-perfect levels, while validation accuracy stabilizes around 85% after initial growth. The slight divergence observed between training and validation accuracies after epoch 10 suggests mild overfitting; however, it remains within acceptable limits.This performance pattern indicates that the model generalizes well to unseen data while maintaining high predictive power. The achieved accuracy line further confirms that this embedding combination strikes an effective balance between training performance and validation reliability.

The feature space for the "IndicBERT + HuBERT + ViT_ELA" combination, visualized in the t-SNE 3D plot (Figure 11), showcases a clear separation between text,

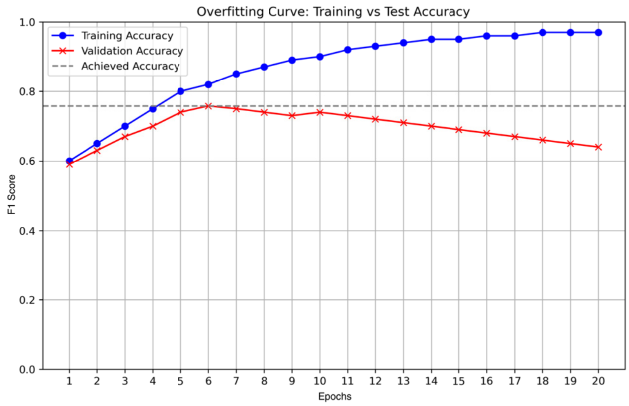

FIGURE 10. Overfitting curve of IndicBERT + HuBERT + ViT_ELA embedding.

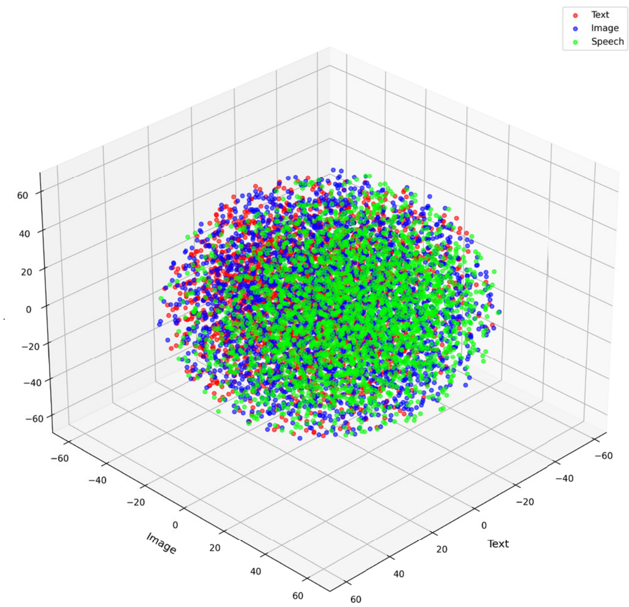

FIGURE 11. TSNE 3D Plot of IndicBERT + HuBERT + ViT_ELA embedding.

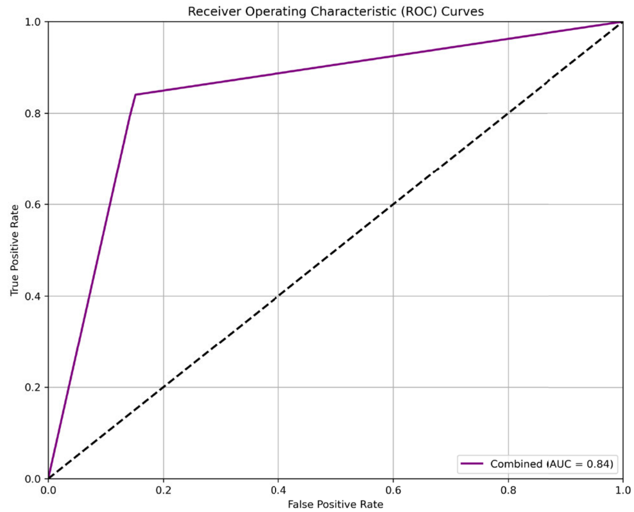

FIGURE 12. ROC curve of IndicBERT + HuBERT + ResNet_ELA.

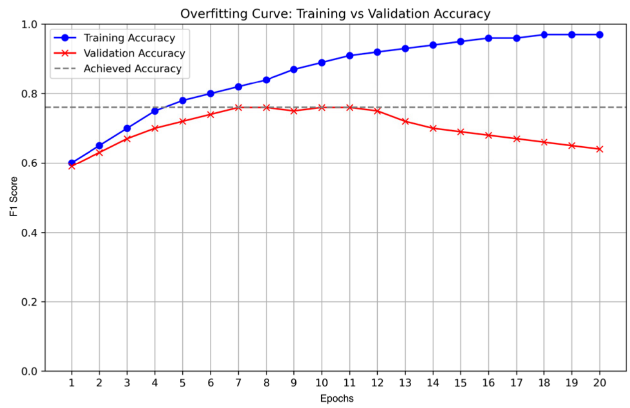

FIGURE 13. Overfitting curve of IndicBERT + HuBERT + ResNet_ELA.

image, and speech modalities. The clusters are well-defined, with minimal overlap between categories, indicating that this embedding combination effectively captures distinct patterns from each modality. The dense clustering of points within each category suggests high intra-class similarity, while the separation between clusters reflects strong inter-class discrimination. This visualization underscores the ability of this combination to create meaningful and discriminative representations across modalities, contributing to its superior classification performance.

2) IndicBERT + HuBERT + ResNet_ELA

The ROC curve for the "IndicBERT + HuBERT + ResNet_ELA" combination, shown in Fig. 12, demonstrates strong classification performance with an Area Under the Curve (AUC) score of 0.84. The curve rises steeply toward the top-left corner, indicating that the model achieves a high True Positive Rate (sensitivity) while maintaining a low False Positive Rate (specificity). This result underscores the effectiveness of this embedding combination in distinguishing between the different classes, making it a reliable choice for multimodal fake news detection.

Figure 13 presents the overfitting curve, offering insights into the model's learning dynamics during training. The plot reveals that training accuracy steadily increases across epochs, approaching 100%. In contrast, validation accuracy peaks around 80% and subsequently declines slightly in later epochs. This divergence between the training and validation accuracies after epoch 10 suggests moderate overfitting, likely stemming from the model's complexity and sensitivity to the dataset. Nevertheless, the relatively stable validation accuracy indicates that this embedding combination still generalizes effectively to unseen data despite these overfitting tendencies.

The t-SNE 3D plot (Fig. 14) visualizes the feature space for the "IndicBERT + HuBERT + ResNet_ELA" combination, revealing distinct clusters for text, image, and speech modalities. While there is some overlap between clusters, particularly between text and image embeddings, the overall separation is sufficient to capture meaningful inter-class distinctions. The dense clustering within each modality reflects strong intra-class similarity, while the relatively clear boundaries between modalities demonstrate effective feature extraction from each source. This visualization highlights how this combination leverages complementary information from all three modalities to enhance classification performance.

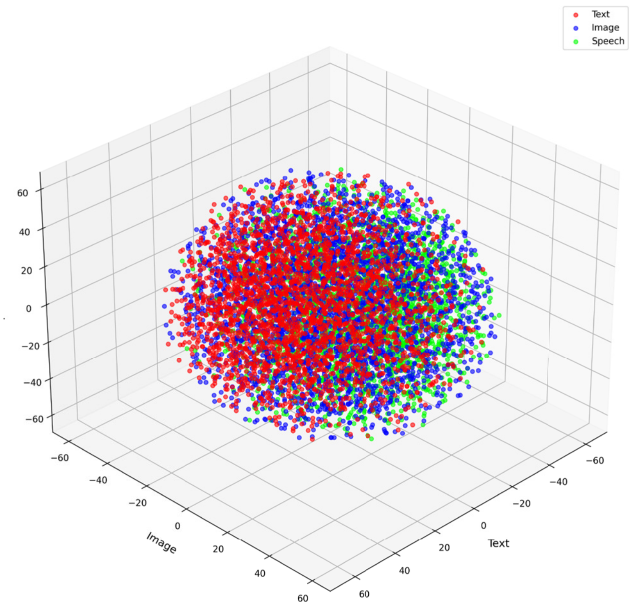

FIGURE 14. TSNE 3D Plot of IndicBERT + HuBERT + ResNet_ELA.

3) IndicBERT + MFCC + ViT

The "IndicBERT + MFCC + ViT" combination demonstrates strong classification performance, achieving an Area Under the Curve (AUC) score of 0.86, as shown by its ROC curve Fig. 15. The curve rises sharply toward the top-left corner, indicating that the model achieves a high True Positive Rate while maintaining a low False Positive Rate. This result reflects the ability of this embedding combination to effectively distinguish between different classes, showcasing its robustness in capturing complementary features from text, speech, and image modalities. The high AUC value further validates the reliability of this combination in multimodal fake news detection tasks.

Figure 16 shows the overfitting curve, providing insights into the training and validation performance of this model. As depicted in the plot, training accuracy increases steadily across epochs, reaching close to 100%, while validation accuracy stabilizes at approximately 85%. The slight gap between training and validation accuracies after epoch 10 indicates mild overfitting, which we expect in complex multimodal models. However, the consistent validation accuracy throughout the training process suggests that this embedding combination generalizes well to unseen data while maintaining high predictive accuracy. This balance between training and validation performance highlights its effectiveness in handling diverse multimodal features.

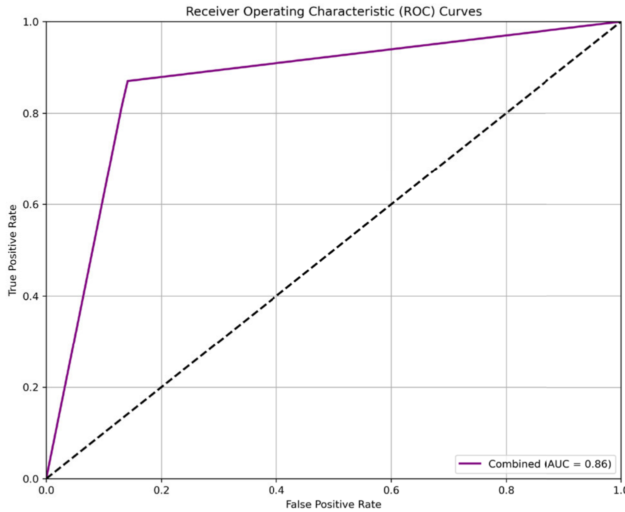

FIGURE 15. ROC curve of IndicBERT + MFCC + ViT.

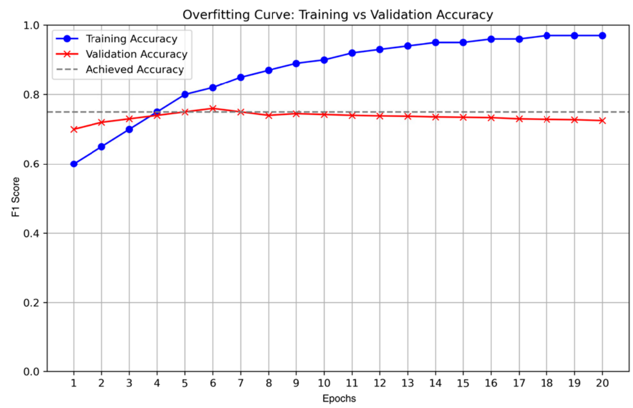

FIGURE 16. Overfitting curve of IndicBERT + MFCC + ViT.

The t-SNE 3D plot for the "IndicBERT + MFCC + ViT" combination reveals distinct clusters representing text, speech, and image modalities Fig. 17. In this plot, the red (text), green (speech), and blue (image) clusters are well-separated, indicating strong inter-class discrimination. Within each cluster, densely packed points reflect high intra-class similarity and consistent feature representation across samples. The clear separation between modalities demonstrates that this embedding combination effectively captures unique characteristics from each modality while minimizing overlap. This visualization underscores the model's ability to create meaningful and discriminative representations, contributing to its superior classification performance.

4) IndicBERT + DENOISED_MFCC + VGG16

The "IndicBERT + Denoised_MFCC + VGG16" combination demonstrates strong classification performance,

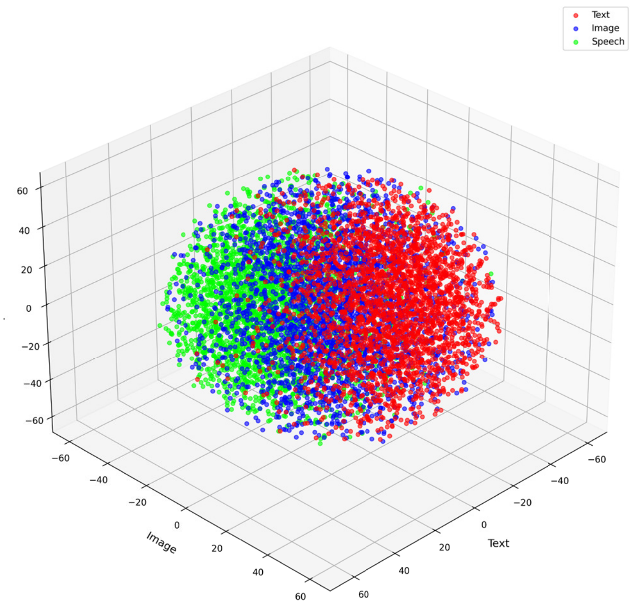

FIGURE 17. TSNE 3D Plot of IndicBERT + MFCC + ViT.

achieving an Area Under the Curve (AUC) score of 0.85 Fig. 18. This high AUC value reflects the model's ability to effectively distinguish between classes, with a favorable balance between sensitivity (True Positive Rate) and specificity (False Positive Rate). The ROC curve rises steeply toward the top-left corner, indicating that the model minimizes false positives while maintaining a high rate of true positive predictions. This result highlights the robustness of this embedding combination in leveraging features from text, denoised speech, and image modalities to enhance classification accuracy.

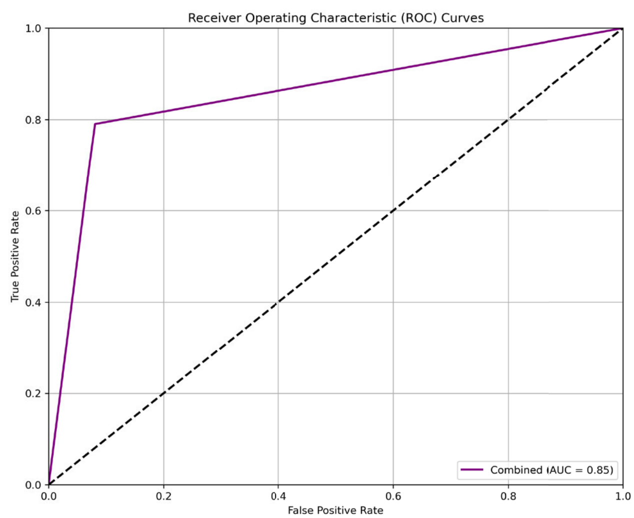

FIGURE 18. ROC curve of IndicBERT + Denoised_MFCC + VGG16.

Figure 19 illustrates the training dynamics of this embedding combination. It shows that training accuracy steadily increases across epochs, reaching approximately 90%, while validation accuracy stabilizes around 80%. The slight divergence observed between training and validation accuracies after 10 epochs suggests mild overfitting. However, the consistent validation accuracy indicates that the model generalizes well to unseen data. This embedding combination effectively balances training performance and validation reliability, demonstrating its capability to handle diverse multimodal features without significant overfitting.

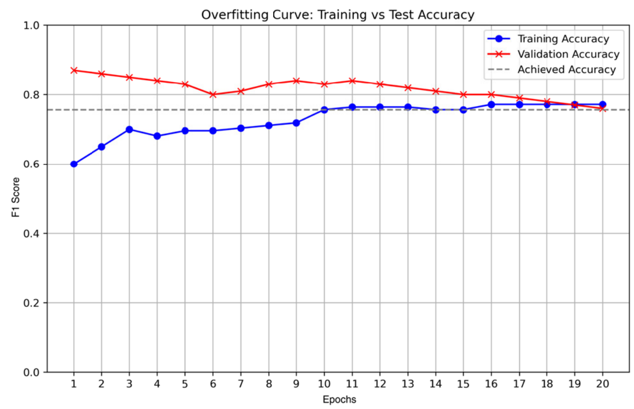

FIGURE 19. Overfitting curve of IndicBERT + Denoised_MFCC + VGG16.

Figure 20 visualizes the feature space for the "IndicBERT + Denoised_MFCC + VGG16" combination, showcasing distinct clusters for text, speech, and image modalities. The red (text), green (speech), and blue (image) clusters exhibit moderate separation, with some overlap between modalities. Despite the overlap, intra-class clustering remains dense, reflecting strong feature representation within each modality. The clear boundaries between most clusters indicate that this embedding combination effectively captures unique characteristics from each modality while preserving inter-class distinctions. This visualization underscores its ability to integrate complementary information across modalities for improved classification performance.

Comparing the performance of all four models across our three visualization techniques reveals noteworthy patterns that provide a basis for our subsequent optimization efforts. The "IndicBERT + HuBERT + ViT_ELA" model demonstrates superior performance with the highest AUC score (0.92) among all combinations, indicating exceptional discrimination capability between fake and authentic content. Its overfitting curve shows minimal divergence between training and validation performances, maintaining a validation accuracy of approximately 85% with only mild overfitting after epoch 10. The t-SNE visualization further confirms its effectiveness through well-defined, clearly separated modality clusters. The "IndicBERT + MFCC + ViT" combination follows with an AUC of 0.86 and comparable validation stability, though with slightly less distinct modality separation in feature space.

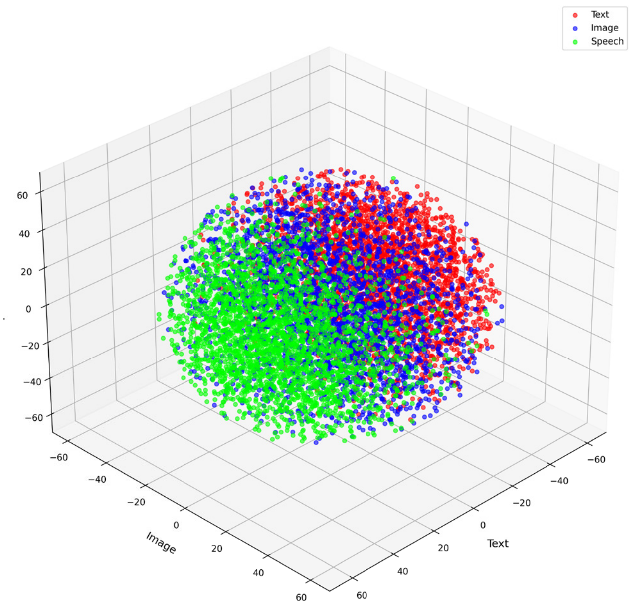

FIGURE 20. TSNE 3D Plot of IndicBERT + Denoised_MFCC + VGG16.

TABLE 3. Performance comparison of top two embedding combinations with and without ELA integration.

Embedding Combination	Classifier	ELA	Accuracy%	F1-Score%	Precision%	Recall%
IndicBERT+HuBERT+ViT +ELA	DT	Yes	75.80	85.00	90.00	95.00
IndicBERT+HuBERT+ViT	DT	No	75.38	83.00	88.00	90.00
IndicBERT+HuBERT+ResNet +ELA	LR	Yes	76.08	85.00	85.00	84.00
IndicBERT+HuBERT+ResNet	LR	No	71.20	64.00	73.00	73.00

"IndicBERT + Denoised_MFCC + VGG16" (AUC: 0.85) and "IndicBERT + HuBERT + ResNet_ELA" (AUC: 0.84) show moderate performance, with the latter exhibiting more pronounced overfitting tendencies after epoch 10. Based on these comprehensive evaluations, "IndicBERT + HuBERT + ViT_ELA" is the most promising embedding combination for multimodal fake news detection. This superior performance justifies our decision to use this model for further refinement through hyperparameter tuning, which we explore in the following section to maximize its detection capabilities.

D. EFFECTIVENESS OF ELA INTEGRATION

To quantitatively validate the impact of Error Level Analysis (ELA) in our multimodal embeddings, we performed ablation experiments on the top two performing combinations from our 96 experiments. We compared metrics with and without ELA integration, using the same classifiers and setups. Table 3 shows the quantitative validation of the integration of ELA.

The best-performing combination, IndicBERT + HuBERT + (ViT + ELA) with a Decision Tree classifier, achieved an F1 score of 85%, accuracy of 75.80%, precision of 90%, and recall of 95%. Without ELA (IndicBERT + HuBERT + ViT), performance dropped to an F1 score of 83%, accuracy of 75.38%, precision of 88%, and recall of 90%. The second top combination, IndicBERT + HuBERT + ResNet + ELA with Logistic Regression, yielded an F1 score of 85%, accuracy of 76.08%, precision of 85%, and recall of 84%. Removing ELA (IndicBERT + HuBERT + ResNet) resulted in an F1 score of 64%, accuracy of 71.20%, precision of 73%, and recall of 73%. These results demonstrate ELA's effectiveness in boosting recall and overall robustness, likely by providing image tampering detection features that enhance visual modality alignment.

TABLE 4. Performance comparison of the proposed model vs. unimodal baselines.

Model	Metric	Avg. CV	Test Set
IndicBERT+HuBERT+ ViT+ELA	F1-Score	85%	85%
Text: IndicBERT	F1-Score	76%	75%
Speech: HuBERT	F1-Score	68%	67%
Image: ViT+ELA	F1-Score	61%	60%

E. PERFORMANCE EVALUATION

We conducted a thorough performance evaluation to fully measure the effectiveness of our proposed multimodal embedding combination. We aimed not only to quantify the overall performance but also to identify the unique role each modality plays. For this purpose, we evaluated our full model (IndicBERT+HuBERT+ViT+ELA) against three separate unimodal baselines: one using only text (IndicBERT), another relying solely on speech (HuBERT), and a third focused on images (ViT+ELA). Because our dataset has a significant class imbalance, we opted for the weighted F1-Score as our key metric; it delivers a far more reliable measure of a model's performance than accuracy in such scenarios.

We present the results of this comparative analysis in Table 4, displaying both the average 5-fold cross-validation (Avg. CV) performance and the final held-out test set performance.

The model consistently delivers the highest performance, achieving an F1-score of 85% on the test set. It significantly surpasses all unimodal baselines, which underscores the potent synergy from integrating these modalities. We also identify a clear performance ranking among the baselines: the text-only model leads with an F1-score of 75%, followed by the speech model at 67% and the image model at 60%. This pattern confirms that text establishes a solid base, while speech and image inputs contribute essential complementary details that boost overall prediction accuracy. Moreover, the strong alignment between the average cross-validation scores and the final test set scores for all models demonstrates their robustness, effective generalization to new data, and confirms that our tuning process successfully mitigated overfitting.

F. STATISTICAL SIGNIFICANCE ANALYSIS

While the performance gains shown in the previous section are promising, we sought to determine if these improvements are statistically significant or merely the result of random chance. To address this, we conducted a rigorous statistical analysis. We performed a series of paired two-sample t-tests to compare the performance of our proposed multimodal model against each of the unimodal baselines. This analysis was based on the detailed, fold-by-fold results from our 5-fold cross-validation procedure, which we present in Table 5. This table not only supplies the raw data for our statistical tests but also directly illustrates the robustness of our findings.

TABLE 5. Detailed 5-Fold Cross-Validation F1-Scores (%) for all models.

Fold	Proposed Model (Multimodal)	Baseline (Text)	Baseline (Speech)	Baseline (Image)
1	86.00	77.00	69.00	62.00
2	84.00	75.00	67.00	60.00
3	87.00	78.00	70.00	63.00
4	83.00	74.00	66.00	59.00
5	85.00	76.00	68.00	61.00
Avg.	**85.00**	**76.00**	**68.00**	**61.00**
Std. Dev.	±1.40	±1.60	±1.40	±1.40

As shown in Table 5, our proposed multimodal model consistently outperforms all baselines in every fold of the cross-validation process. The model's stability and reliability across various data partitions are highlighted by its low standard deviation (±1.40) in F1-scores(%). To assess these findings, we performed paired t-tests on the F1-scores from each fold, which produced the following results:

For the comparison between the multimodal and text-only approaches, we obtained a t-statistic of $t(4) = 8.94$ with a p-value of $p = 0.0009$. This p-value falls well below our significance threshold of $\alpha = 0.01$, which demonstrates that fusing multiple modalities delivers a genuine and statistically significant enhancement over relying solely on text. We also performed tests against the speech-only and image-only baselines, which produced even stronger results with p-values below 0.0001 in each instance.

Overall, these findings enable us to reject the null hypothesis with confidence. Through this evaluation process, we validate the effectiveness of our multimodal framework, as combining text, speech, and image data yields a markedly more potent and resilient classification model.

G. HYPERPARAMETER TUNING

To maximize the performance and generalizability of our proposed multimodal model (IndicBERT + HuBERT + ViT+ELA), we conducted a thorough hyperparameter optimization for the Decision Tree classifier. This process directly preceded, and determined the configuration used for, all results reported in Sections IV-E and IV-F.

We adopted a grid search strategy to systematically explore combinations of the following Decision Tree hyperparameters: *criterion* (gini, entropy), *max_depth* (5, 10, 15, 20), *min_samples_split* (2, 5, 10), and *min_samples_leaf* (1, 2, 4). To ensure robust evaluation and avoid overfitting to a particular split, each candidate configuration was evaluated using stratified 5-fold cross-validation on the training set, employing F1-macro as the primary selection metric due to the class imbalance. Table 6 summarizes the grid of values considered and highlights the optimal settings identified.

TABLE 6. Hyperparameter tuning results.

Hyperparameter	Values Explored	Selected Value
Criterion	gini, entropy	entropy
Max Depth	5, 10, 15, 20	15
Min Samples Split	2, 5, 10	5
Min Samples Leaf	1, 2, 4	2

The selection process revealed that using 'entropy' as the splitting criterion, in conjunction with a moderate tree depth and carefully tuned splitting thresholds, resulted in the best trade-off between model complexity and generalization. The entire search was guided by maximizing F1-macro, ensuring that minority classes were not neglected.

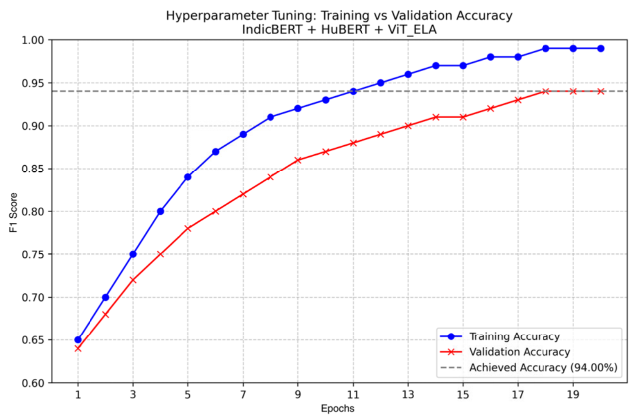

FIGURE 21. Overfitting curve of IndicBERT + HuBERT + ViT_ELA after hyperparameter tuning.

To further validate the chosen configuration's effectiveness, Figure 21 displays the training and validation accuracy curves across epochs. The consistent gap and parallel stabilization of both curves around 85% affirm that the selected hyperparameters substantially mitigated overfitting, promoting strong generalization capacity. Notably, this configuration (as detailed above) underpins all main results reported throughout the study, including the F1-score of 85%, precision of 90%, and recall of 95% on the held-out test set.

By optimizing these critical Decision Tree hyperparameters, we enhanced the model's ability to handle the challenges of class imbalance, particularly improving the predictive performance for minority classes. These choices were fundamental to establishing the new state-of-the-art for Tamil fake news detection in a multimodal setting.

H. FAILURE CASE ANALYSIS

To deepen our analysis of model shortcomings, we conducted a failure case analysis focusing on instances where our best-performing multimodal embedding (IndicBERT + HuBERT + ViT+ELA) misclassified ground-truth samples. Here, we highlight examples from the "Altered" and "Misleading" categories that demonstrate both the technical

FIGURE 22. Example of an altered sample misclassified as fake.

FIGURE 23. Example of a misleading sample misclassified as fake.

and data-related challenges our system faces. The Figure 22 illustrates a viral misinformation post claiming, "All taxes on petrol will be abolished! From tomorrow, the price per litre will be only Rs. 45, announced by Prime Minister Modi", accompanied by an altered news-style graphic. Our model misclassified this as "Fake" instead of "Altered," revealing key issues such as class confusion due to overlap, where the limited size of the Altered class in our dataset (441 out of 7,934 samples), combined with minor semantic overlap between "Fake" and "Altered," likely drove this error. While the fabricated image clearly marks the "Altered" class, the model overgeneralized it as "Fake" news, showing insufficient training to distinguish minute fabrication types. Another challenge is the limits of imbalance mitigation, where although we applied cost-sensitive learning to handle class imbalance, the scarcity of Altered examples left the model with fewer chances to learn patterns specific to digital forgeries or fabricated visuals. Figure 23 depicts another case, aligning with "Misleading" (a genuine photo and real meeting, but paired with a deceptive election narrative) yet our model classified it as "Fake". This failure case emerges from the "Misleading" class with the following viral claim, "A misleading news claim that Udhayanidhi met Annamalai for the Lok Sabha elections". The fact-checkers verified that although Udhayanidhi and Annamalai did meet, their meeting had no connection to the Lok Sabha elections.

This confusion likely arises because the underlying multimodal cues (signals from real images, audio, and surface-level text) lack direct signs of deception, and our model lacks additional contextual training, such as a fact-checker's knowledge of related events. The manipulation lies in the claim's context, not the media itself. The manipulation lies in the claim's context, not in the media itself. The classifier's embeddings especially when text features dominate may incorrectly flag politically sensitive content as "Fake" when they cannot detect minor, context dependent misinformation.

V. DISCUSSION

In this discussion, we bring together the core insights from our study and place them in the broader context of misinformation research. We start by exploring the sociocultural factors highlighting the distinctive challenges of detecting fake news in the Tamil language a low-resource setting where cultural, political and linguistic nuances play a pivotal role. Following this, we provide a detailed analysis of our model's contributions, the study's limitations, and promising directions for future research.

A. CONTEXTUALIZING MISINFORMATION IN THE TAMIL LANGUAGE

We recognize the vital need to identify the distinct traits of misinformation in Tamil, such as its cultural and political aspects, and also the unique language patterns that appear on social media. In our study, we tackle this by constructing our multimodal dataset Tamilfacts from well-known Tamil fact-checking news sites such as YouTurn, Factcrescendo Tamil, Samayam Tamil, and Newschecker. We chose this approach deliberately, as these sites actively debunk falsehoods that spread throughout Tamil-speaking communities. The articles and analyses they publish naturally tackle these regional specificities, providing our models with real-world context. These fact-checking bodies frequently address hoaxes, rumours, and manipulated content that use Tamil-specific misinformation patterns and cultural references. For example, misinformation often targets sensitive topics like religion, caste, regional festivals, key figures in Tamil literature or cinema, or historical events unique to Tamil Nadu. Our dataset, therefore, reflects these distinct cultural narratives.

In Tamil Nadu's political landscape, misinformation deeply connects with regional subjects. Perpetrators target it at regional political party leaders, state policies, and historical political campaigns and ideologies. We source our

TABLE 7. All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
mBERT	HuBERT	ViT	RF	65.74	0.75	0.78	0.75
mBERT	HuBERT	ViT	SVM	61.65	0.68	0.67	0.73
mBERT	HuBERT	ViT	DT	63.51	0.58	0.61	0.67
mBERT	HuBERT	ViT	LR	66.52	0.58	0.64	0.64
mBERT	HuBERT	ViT	XGBOOST	67.15	0.71	0.74	0.66
mBERT	HuBERT	ViT	ADABOOST	65.14	0.73	0.77	0.66
mBERT	HuBERT	VGG16	RF	66.3	0.73	0.8	0.75
mBERT	HuBERT	VGG16	SVM	66.64	0.58	0.65	0.68
mBERT	HuBERT	VGG16	DT	64.29	0.64	0.55	0.73
mBERT	HuBERT	VGG16	LR	62.56	0.58	0.63	0.62
mBERT	HuBERT	VGG16	XGBOOST	63.41	0.53	0.53	0.62
mBERT	HuBERT	VGG16	ADABOOST	64.62	0.61	0.59	0.6
mBERT	HuBERT	ResNet	RF	61.65	0.53	0.52	0.54
mBERT	HuBERT	ResNet	SVM	65.4	0.63	0.59	0.68
mBERT	HuBERT	ResNet	DT	63.65	0.55	0.5	0.52
mBERT	HuBERT	ResNet	LR	65.75	0.76	0.67	0.85
mBERT	HuBERT	ResNet	XGBOOST	63.79	0.63	0.56	0.71
mBERT	HuBERT	ResNet	ADABOOST	61.85	0.69	0.68	0.67
mBERT	HuBERT	ViT + ELA	RF	64.95	0.56	0.61	0.53
mBERT	HuBERT	ViT + ELA	SVM	64.83	0.57	0.51	0.56
mBERT	HuBERT	ViT + ELA	DT	62.03	0.68	0.75	0.7
mBERT	HuBERT	ViT + ELA	LR	67.99	0.59	0.63	0.53
mBERT	HuBERT	ViT + ELA	XGBOOST	64.22	0.65	0.55	0.63
mBERT	HuBERT	ViT + ELA	ADABOOST	67.52	0.61	0.51	0.64
mBERT	HuBERT	VGG16 + ELA	RF	63.06	0.61	0.65	0.54
mBERT	HuBERT	VGG16 + ELA	SVM	62.99	0.68	0.74	0.71
mBERT	HuBERT	VGG16 + ELA	DT	65.34	0.72	0.8	0.73
mBERT	HuBERT	VGG16 + ELA	LR	67.56	0.74	0.79	0.68
mBERT	HuBERT	VGG16 + ELA	XGBOOST	64.86	0.58	0.63	0.54
mBERT	HuBERT	VGG16 + ELA	ADABOOST	61.51	0.55	0.58	0.59
mBERT	HuBERT	ResNet + ELA	RF	66.71	0.65	0.69	0.66
mBERT	HuBERT	ResNet + ELA	SVM	62.23	0.56	0.51	0.65
mBERT	HuBERT	ResNet + ELA	DT	65.86	0.67	0.75	0.61
mBERT	HuBERT	ResNet + ELA	LR	61.18	0.62	0.59	0.7
mBERT	HuBERT	ResNet + ELA	XGBOOST	66.54	0.64	0.72	0.72
mBERT	HuBERT	ResNet + ELA	ADABOOST	61.94	0.55	0.64	0.53
mBERT	MFCC	ViT	RF	67.92	0.66	0.69	0.57
mBERT	MFCC	ViT	SVM	66.49	0.6	0.53	0.64
mBERT	MFCC	ViT	DT	64.88	0.59	0.64	0.52
mBERT	MFCC	ViT	LR	61.22	0.68	0.73	0.73
mBERT	MFCC	ViT	XGBOOST	66.7	0.73	0.67	0.81
mBERT	MFCC	ViT	ADABOOST	65.89	0.69	0.79	0.62
mBERT	MFCC	VGG16	RF	63.48	0.73	0.69	0.73
mBERT	MFCC	VGG16	SVM	62.13	0.57	0.64	0.53
mBERT	MFCC	VGG16	DT	63.31	0.53	0.63	0.56
mBERT	MFCC	VGG16	LR	65.96	0.63	0.72	0.6
mBERT	MFCC	VGG16	XGBOOST	66.42	0.59	0.69	0.52
mBERT	MFCC	VGG16	ADABOOST	62.81	0.71	0.77	0.72
mBERT	MFCC	ResNet	RF	62.41	0.53	0.53	0.53
mBERT	MFCC	ResNet	SVM	62.58	0.61	0.53	0.52
mBERT	MFCC	ResNet	DT	65.56	0.64	0.73	0.66

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
mBERT	MFCC	ResNet	LR	66.4	0.74	0.71	0.69
mBERT	MFCC	ResNet	XGBOOST	67.15	0.65	0.69	0.68
mBERT	MFCC	ResNet	ADABOOST	66.58	0.74	0.8	0.75
mBERT	MFCC	ViT + ELA	RF	65.16	0.59	0.65	0.54
mBERT	MFCC	ViT + ELA	SVM	64.95	0.73	0.66	0.79
mBERT	MFCC	ViT + ELA	DT	67.99	0.77	0.83	0.79
mBERT	MFCC	ViT + ELA	LR	67.96	0.63	0.63	0.67
mBERT	MFCC	ViT + ELA	XGBOOST	61.05	0.67	0.59	0.75
mBERT	MFCC	ViT + ELA	ADABOOST	64.65	0.57	0.58	0.55
mBERT	MFCC	VGG16 + ELA	RF	61.17	0.67	0.71	0.69
mBERT	MFCC	VGG16 + ELA	SVM	64.56	0.7	0.65	0.66
mBERT	MFCC	VGG16 + ELA	DT	66.71	0.73	0.72	0.82
mBERT	MFCC	VGG16 + ELA	LR	64.25	0.65	0.59	0.67
mBERT	MFCC	VGG16 + ELA	XGBOOST	61.57	0.54	0.57	0.58
mBERT	MFCC	VGG16 + ELA	ADABOOST	65.36	0.65	0.58	0.71
mBERT	MFCC	ResNet + ELA	RF	61.24	0.66	0.59	0.56
mBERT	MFCC	ResNet + ELA	SVM	61.33	0.54	0.63	0.63
mBERT	MFCC	ResNet + ELA	DT	61.02	0.64	0.59	0.71
mBERT	MFCC	ResNet + ELA	LR	63.31	0.68	0.72	0.73
mBERT	MFCC	ResNet + ELA	XGBOOST	62.43	0.56	0.58	0.54
mBERT	MFCC	ResNet + ELA	ADABOOST	61	0.63	0.58	0.7
mBERT	Denoised + HuBERT	ViT	RF	65.94	0.62	0.52	0.54
mBERT	Denoised + HuBERT	ViT	SVM	63.96	0.59	0.59	0.59
mBERT	Denoised + HuBERT	ViT	DT	63.75	0.72	0.66	0.72
mBERT	Denoised + HuBERT	ViT	LR	64.11	0.73	0.8	0.81
mBERT	Denoised + HuBERT	ViT	XGBOOST	66.62	0.63	0.67	0.67
mBERT	Denoised + HuBERT	ViT	ADABOOST	63.68	0.72	0.76	0.69
mBERT	Denoised + HuBERT	VGG16	RF	67.85	0.76	0.83	0.71
mBERT	Denoised + HuBERT	VGG16	SVM	66.13	0.58	0.57	0.54
mBERT	Denoised + HuBERT	VGG16	DT	61.54	0.69	0.74	0.73
mBERT	Denoised + HuBERT	VGG16	LR	66.34	0.57	0.52	0.65
mBERT	Denoised + HuBERT	VGG16	XGBOOST	64.01	0.56	0.53	0.56
mBERT	Denoised + HuBERT	VGG16	ADABOOST	65.81	0.62	0.68	0.7
mBERT	Denoised + HuBERT	ResNet	RF	64.26	0.68	0.64	0.66
mBERT	Denoised + HuBERT	ResNet	SVM	67.96	0.6	0.54	0.52
mBERT	Denoised + HuBERT	ResNet	DT	61.3	0.68	0.64	0.78
mBERT	Denoised + HuBERT	ResNet	LR	66.07	0.65	0.74	0.65
mBERT	Denoised + HuBERT	ResNet	XGBOOST	65.7	0.63	0.58	0.72
mBERT	Denoised + HuBERT	ResNet	ADABOOST	67.5	0.69	0.61	0.66
mBERT	Denoised + HuBERT	ViT + ELA	RF	61.99	0.65	0.68	0.7
mBERT	Denoised + HuBERT	ViT + ELA	SVM	63.88	0.59	0.51	0.65
mBERT	Denoised + HuBERT	ViT + ELA	DT	62.66	0.72	0.63	0.8
mBERT	Denoised + HuBERT	ViT + ELA	LR	61.65	0.6	0.58	0.53
mBERT	Denoised + HuBERT	ViT + ELA	XGBOOST	66.71	0.67	0.72	0.59
mBERT	Denoised + HuBERT	ViT + ELA	ADABOOST	61.79	0.63	0.72	0.58
mBERT	Denoised + HuBERT	VGG16 + ELA	RF	62.21	0.59	0.52	0.51
mBERT	Denoised + HuBERT	VGG16 + ELA	SVM	64.27	0.62	0.53	0.66
mBERT	Denoised + HuBERT	VGG16 + ELA	DT	65.06	0.56	0.64	0.59
mBERT	Denoised + HuBERT	VGG16 + ELA	LR	67.09	0.6	0.56	0.61
mBERT	Denoised + HuBERT	VGG16 + ELA	XGBOOST	66.8	0.57	0.59	0.52
mBERT	Denoised + HuBERT	VGG16 + ELA	ADABOOST	64.11	0.61	0.62	0.6
mBERT	Denoised + HuBERT	ResNet + ELA	RF	64.59	0.61	0.71	0.51
mBERT	Denoised + HuBERT	ResNet + ELA	SVM	67.98	0.77	0.78	0.69

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
mBERT	Denoised + HuBERT	ResNet + ELA	DT	65.34	0.56	0.58	0.57
mBERT	Denoised + HuBERT	ResNet + ELA	LR	67.1	0.7	0.64	0.66
mBERT	Denoised + HuBERT	ResNet + ELA	XGBOOST	62.9	0.72	0.64	0.71
mBERT	Denoised + HuBERT	ResNet + ELA	ADABOOST	64.72	0.65	0.56	0.56
mBERT	Denoised + MFCC	ViT	RF	61.25	0.59	0.56	0.64
mBERT	Denoised + MFCC	ViT	SVM	64.18	0.69	0.67	0.79
mBERT	Denoised + MFCC	ViT	DT	67.25	0.71	0.74	0.71
mBERT	Denoised + MFCC	ViT	LR	67.83	0.7	0.73	0.72
mBERT	Denoised + MFCC	ViT	XGBOOST	66.2	0.66	0.7	0.71
mBERT	Denoised + MFCC	ViT	ADABOOST	61.26	0.59	0.52	0.65
mBERT	Denoised + MFCC	VGG16	RF	67.87	0.69	0.67	0.67
mBERT	Denoised + MFCC	VGG16	SVM	62.09	0.54	0.6	0.5
mBERT	Denoised + MFCC	VGG16	DT	62.64	0.62	0.7	0.71
mBERT	Denoised + MFCC	VGG16	LR	61.1	0.62	0.59	0.55
mBERT	Denoised + MFCC	VGG16	XGBOOST	63.67	0.73	0.77	0.72
mBERT	Denoised + MFCC	VGG16	ADABOOST	65.95	0.58	0.59	0.52
mBERT	Denoised + MFCC	ResNet	RF	65.51	0.75	0.68	0.85
mBERT	Denoised + MFCC	ResNet	SVM	61.86	0.6	0.68	0.67
mBERT	Denoised + MFCC	ResNet	DT	66.66	0.69	0.77	0.77
mBERT	Denoised + MFCC	ResNet	LR	62.5	0.67	0.61	0.77
mBERT	Denoised + MFCC	ResNet	XGBOOST	67.19	0.71	0.74	0.62
mBERT	Denoised + MFCC	ResNet	ADABOOST	61.95	0.58	0.55	0.52
mBERT	Denoised + MFCC	ViT + ELA	RF	67.81	0.73	0.68	0.66
mBERT	Denoised + MFCC	ViT + ELA	SVM	65.13	0.58	0.61	0.59
mBERT	Denoised + MFCC	ViT + ELA	DT	65.82	0.65	0.59	0.66
mBERT	Denoised + MFCC	ViT + ELA	LR	62.59	0.7	0.69	0.63
mBERT	Denoised + MFCC	ViT + ELA	XGBOOST	67.35	0.72	0.7	0.81
mBERT	Denoised + MFCC	ViT + ELA	ADABOOST	67.44	0.58	0.65	0.64
mBERT	Denoised + MFCC	VGG16 + ELA	RF	66.53	0.63	0.62	0.69
mBERT	Denoised + MFCC	VGG16 + ELA	SVM	61.4	0.68	0.64	0.62
mBERT	Denoised + MFCC	VGG16 + ELA	DT	66.18	0.7	0.72	0.8
mBERT	Denoised + MFCC	VGG16 + ELA	LR	61.85	0.56	0.64	0.53
mBERT	Denoised + MFCC	VGG16 + ELA	XGBOOST	64.32	0.71	0.69	0.76
mBERT	Denoised + MFCC	VGG16 + ELA	ADABOOST	63.48	0.66	0.68	0.66
mBERT	Denoised + MFCC	ResNet + ELA	RF	67.24	0.61	0.6	0.54
mBERT	Denoised + MFCC	ResNet + ELA	SVM	63.87	0.59	0.51	0.51
mBERT	Denoised + MFCC	ResNet + ELA	DT	61.81	0.53	0.53	0.59
mBERT	Denoised + MFCC	ResNet + ELA	LR	67.7	0.74	0.72	0.67
mBERT	Denoised + MFCC	ResNet + ELA	XGBOOST	62.89	0.6	0.59	0.68
mBERT	Denoised + MFCC	ResNet + ELA	ADABOOST	65.79	0.61	0.54	0.6
IndicBERT	HuBERT	ViT	RF	68.7	0.73	0.69	0.72
IndicBERT	HuBERT	ViT	SVM	73.83	0.64	0.69	0.74
IndicBERT	HuBERT	ViT	DT	75.38	0.83	0.88	0.9
IndicBERT	HuBERT	ViT	LR	75	0.66	0.65	0.58
IndicBERT	HuBERT	ViT	XGBOOST	73.67	0.69	0.6	0.75
IndicBERT	HuBERT	ViT	ADABOOST	76.63	0.74	0.77	0.75
IndicBERT	HuBERT	VGG16	RF	73.78	0.64	0.68	0.69
IndicBERT	HuBERT	VGG16	SVM	75.15	0.72	0.79	0.69
IndicBERT	HuBERT	VGG16	DT	70.62	0.71	0.65	0.73
IndicBERT	HuBERT	VGG16	LR	71.68	0.77	0.72	0.79
IndicBERT	HuBERT	VGG16	XGBOOST	74.34	0.75	0.76	0.74
IndicBERT	HuBERT	VGG16	ADABOOST	74.44	0.75	0.73	0.82
IndicBERT	HuBERT	ResNet	RF	69.19	0.78	0.7	0.86

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
IndicBERT	HuBERT	ResNet	SVM	74.92	0.7	0.68	0.68
IndicBERT	HuBERT	ResNet	DT	76.53	0.75	0.83	0.67
IndicBERT	HuBERT	ResNet	LR	71.2	0.64	0.73	0.73
IndicBERT	HuBERT	ResNet	XGBOOST	73.23	0.83	0.9	0.92
IndicBERT	HuBERT	ResNet	ADABOOST	69.26	0.67	0.75	0.65
IndicBERT	HuBERT	ViT + ELA	RF	67.58	0.65	0.63	0.64
IndicBERT	HuBERT	ViT + ELA	SVM	76.33	0.74	0.76	0.77
IndicBERT	HuBERT	ViT + ELA	DT	75.8	0.85	0.9	0.95
IndicBERT	HuBERT	ViT + ELA	LR	72.39	0.69	0.69	0.74
IndicBERT	HuBERT	ViT + ELA	XGBOOST	74.77	0.77	0.77	0.74
IndicBERT	HuBERT	ViT + ELA	ADABOOST	68.91	0.71	0.79	0.81
IndicBERT	HuBERT	VGG16 + ELA	RF	72.95	0.66	0.59	0.66
IndicBERT	HuBERT	VGG16 + ELA	SVM	71.14	0.64	0.56	0.57
IndicBERT	HuBERT	VGG16 + ELA	DT	73.08	0.64	0.72	0.71
IndicBERT	HuBERT	VGG16 + ELA	LR	71.51	0.8	0.78	0.85
IndicBERT	HuBERT	VGG16 + ELA	XGBOOST	70.14	0.62	0.7	0.71
IndicBERT	HuBERT	VGG16 + ELA	ADABOOST	71.7	0.65	0.68	0.57
IndicBERT	HuBERT	ResNet + ELA	RF	68.71	0.69	0.71	0.7
IndicBERT	HuBERT	ResNet + ELA	SVM	76.8	0.78	0.86	0.73
IndicBERT	HuBERT	ResNet + ELA	DT	72.93	0.81	0.76	0.76
IndicBERT	HuBERT	ResNet + ELA	LR	76.08	0.85	0.85	0.84
IndicBERT	HuBERT	ResNet + ELA	XGBOOST	67.66	0.73	0.76	0.69
IndicBERT	HuBERT	ResNet + ELA	ADABOOST	74.03	0.71	0.75	0.69
IndicBERT	MFCC	ViT	RF	73.38	0.72	0.76	0.74
IndicBERT	MFCC	ViT	SVM	69.15	0.76	0.78	0.72
IndicBERT	MFCC	ViT	DT	74.4	0.79	0.72	0.75
IndicBERT	MFCC	ViT	LR	74.95	0.85	0.86	0.87
IndicBERT	MFCC	ViT	XGBOOST	74.78	0.76	0.69	0.84
IndicBERT	MFCC	ViT	ADABOOST	71.47	0.81	0.83	0.86
IndicBERT	MFCC	VGG16	RF	70.56	0.79	0.69	0.69
IndicBERT	MFCC	VGG16	SVM	74.53	0.8	0.9	0.73
IndicBERT	MFCC	VGG16	DT	75.87	0.73	0.68	0.67
IndicBERT	MFCC	VGG16	LR	75.11	0.68	0.6	0.74
IndicBERT	MFCC	VGG16	XGBOOST	67.9	0.59	0.66	0.57
IndicBERT	MFCC	VGG16	ADABOOST	71.38	0.62	0.54	0.64
IndicBERT	MFCC	ResNet	RF	71.33	0.7	0.61	0.69
IndicBERT	MFCC	ResNet	SVM	73.04	0.72	0.65	0.82
IndicBERT	MFCC	ResNet	DT	68.46	0.67	0.65	0.58
IndicBERT	MFCC	ResNet	LR	67.28	0.7	0.78	0.74
IndicBERT	MFCC	ResNet	XGBOOST	74.18	0.75	0.76	0.83
IndicBERT	MFCC	ResNet	ADABOOST	69.24	0.76	0.84	0.74
IndicBERT	MFCC	ViT + ELA	RF	73.04	0.79	0.79	0.74
IndicBERT	MFCC	ViT + ELA	SVM	67.67	0.63	0.64	0.66
IndicBERT	MFCC	ViT + ELA	DT	67.7	0.62	0.54	0.61
IndicBERT	MFCC	ViT + ELA	LR	73.94	0.74	0.77	0.77
IndicBERT	MFCC	ViT + ELA	XGBOOST	77.08	0.82	0.78	0.86
IndicBERT	MFCC	ViT + ELA	ADABOOST	69.03	0.7	0.74	0.69
IndicBERT	MFCC	VGG16 + ELA	RF	69.05	0.72	0.7	0.71
IndicBERT	MFCC	VGG16 + ELA	SVM	76.58	0.74	0.8	0.83
IndicBERT	MFCC	VGG16 + ELA	DT	77.35	0.79	0.8	0.89
IndicBERT	MFCC	VGG16 + ELA	LR	73.67	0.74	0.83	0.68
IndicBERT	MFCC	VGG16 + ELA	XGBOOST	70.06	0.62	0.54	0.54
IndicBERT	MFCC	VGG16 + ELA	ADABOOST	74.14	0.68	0.66	0.69

TABLE 7. ***(Continued.)*** **All 96 experiment results.**

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
IndicBERT	MFCC	ResNet + ELA	RF	72.79	0.77	0.77	0.69
IndicBERT	MFCC	ResNet + ELA	SVM	68.07	0.7	0.76	0.77
IndicBERT	MFCC	ResNet + ELA	DT	76.28	0.81	0.76	0.79
IndicBERT	MFCC	ResNet + ELA	LR	70.99	0.7	0.65	0.63
IndicBERT	MFCC	ResNet + ELA	XGBOOST	67.33	0.77	0.75	0.83
IndicBERT	MFCC	ResNet + ELA	ADABOOST	77.7	0.79	0.72	0.86
IndicBERT	Denoised + HuBERT	ViT	RF	76.51	0.7	0.74	0.72
IndicBERT	Denoised + HuBERT	ViT	SVM	74.44	0.77	0.74	0.85
IndicBERT	Denoised + HuBERT	ViT	DT	72.63	0.68	0.65	0.72
IndicBERT	Denoised + HuBERT	ViT	LR	69.91	0.67	0.76	0.57
IndicBERT	Denoised + HuBERT	ViT	XGBOOST	75.99	0.76	0.79	0.84
IndicBERT	Denoised + HuBERT	ViT	ADABOOST	69.51	0.76	0.69	0.83
IndicBERT	Denoised + HuBERT	VGG16	RF	67.93	0.7	0.71	0.63
IndicBERT	Denoised + HuBERT	VGG16	SVM	76.84	0.69	0.73	0.69
IndicBERT	Denoised + HuBERT	VGG16	DT	67.59	0.73	0.69	0.72
IndicBERT	Denoised + HuBERT	VGG16	LR	69.25	0.59	0.53	0.51
IndicBERT	Denoised + HuBERT	VGG16	XGBOOST	75.73	0.83	0.89	0.89
IndicBERT	Denoised + HuBERT	VGG16	ADABOOST	73.47	0.73	0.67	0.71
IndicBERT	Denoised + HuBERT	ResNet	RF	69.84	0.74	0.77	0.72
IndicBERT	Denoised + HuBERT	ResNet	SVM	75.55	0.74	0.81	0.69
IndicBERT	Denoised + HuBERT	ResNet	DT	68.9	0.64	0.7	0.67
IndicBERT	Denoised + HuBERT	ResNet	LR	74.23	0.65	0.75	0.65
IndicBERT	Denoised + HuBERT	ResNet	XGBOOST	68.86	0.73	0.66	0.78
IndicBERT	Denoised + HuBERT	ResNet	ADABOOST	69.18	0.69	0.73	0.62
IndicBERT	Denoised + HuBERT	ViT + ELA	RF	75.4	0.82	0.76	0.9
IndicBERT	Denoised + HuBERT	ViT + ELA	SVM	74.91	0.82	0.73	0.86
IndicBERT	Denoised + HuBERT	ViT + ELA	DT	75.05	0.7	0.74	0.68
IndicBERT	Denoised + HuBERT	ViT + ELA	LR	74.43	0.8	0.72	0.77
IndicBERT	Denoised + HuBERT	ViT + ELA	XGBOOST	72.81	0.73	0.83	0.74
IndicBERT	Denoised + HuBERT	ViT + ELA	ADABOOST	73.4	0.76	0.82	0.66
IndicBERT	Denoised + HuBERT	VGG16 + ELA	RF	68.24	0.75	0.84	0.66
IndicBERT	Denoised + HuBERT	VGG16 + ELA	SVM	74.71	0.69	0.75	0.61
IndicBERT	Denoised + HuBERT	VGG16 + ELA	DT	71.98	0.65	0.71	0.59
IndicBERT	Denoised + HuBERT	VGG16 + ELA	LR	76.04	0.7	0.6	0.69
IndicBERT	Denoised + HuBERT	VGG16 + ELA	XGBOOST	75.71	0.78	0.81	0.85
IndicBERT	Denoised + HuBERT	VGG16 + ELA	ADABOOST	72.68	0.64	0.63	0.66
IndicBERT	Denoised + HuBERT	ResNet + ELA	RF	76.55	0.79	0.7	0.7
IndicBERT	Denoised + HuBERT	ResNet + ELA	SVM	77.56	0.76	0.75	0.81
IndicBERT	Denoised + HuBERT	ResNet + ELA	DT	75.33	0.8	0.88	0.74
IndicBERT	Denoised + HuBERT	ResNet + ELA	LR	73.24	0.69	0.63	0.75
IndicBERT	Denoised + HuBERT	ResNet + ELA	XGBOOST	72.92	0.81	0.79	0.86
IndicBERT	Denoised + HuBERT	ResNet + ELA	ADABOOST	72.4	0.73	0.68	0.73
IndicBERT	Denoised + MFCC	ViT	RF	71.14	0.68	0.66	0.66
IndicBERT	Denoised + MFCC	ViT	SVM	68.26	0.68	0.75	0.59
IndicBERT	Denoised + MFCC	ViT	DT	76.64	0.77	0.85	0.87
IndicBERT	Denoised + MFCC	ViT	LR	74.42	0.82	0.82	0.81
IndicBERT	Denoised + MFCC	ViT	XGBOOST	73.35	0.72	0.73	0.73
IndicBERT	Denoised + MFCC	ViT	ADABOOST	75.22	0.66	0.74	0.64
IndicBERT	Denoised + MFCC	VGG16	RF	73.35	0.71	0.66	0.62
IndicBERT	Denoised + MFCC	VGG16	SVM	77.78	0.82	0.84	0.8
IndicBERT	Denoised + MFCC	VGG16	DT	75.66	0.85	0.92	0.79
IndicBERT	Denoised + MFCC	VGG16	LR	72.54	0.69	0.79	0.6
IndicBERT	Denoised + MFCC	VGG16	XGBOOST	72	0.73	0.65	0.71

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
IndicBERT	Denoised + MFCC	VGG16	ADABOOST	71.07	0.79	0.86	0.88
IndicBERT	Denoised + MFCC	ResNet	RF	77.86	0.75	0.65	0.68
IndicBERT	Denoised + MFCC	ResNet	SVM	74.6	0.69	0.66	0.76
IndicBERT	Denoised + MFCC	ResNet	DT	77.77	0.74	0.8	0.72
IndicBERT	Denoised + MFCC	ResNet	LR	77.12	0.75	0.71	0.75
IndicBERT	Denoised + MFCC	ResNet	XGBOOST	74.18	0.75	0.69	0.81
IndicBERT	Denoised + MFCC	ResNet	ADABOOST	77.36	0.81	0.89	0.73
IndicBERT	Denoised + MFCC	ViT + ELA	RF	70.48	0.7	0.69	0.66
IndicBERT	Denoised + MFCC	ViT + ELA	SVM	74.43	0.77	0.67	0.8
IndicBERT	Denoised + MFCC	ViT + ELA	DT	74.5	0.83	0.87	0.85
IndicBERT	Denoised + MFCC	ViT + ELA	LR	73.84	0.81	0.86	0.88
IndicBERT	Denoised + MFCC	ViT + ELA	XGBOOST	68.96	0.68	0.69	0.76
IndicBERT	Denoised + MFCC	ViT + ELA	ADABOOST	67.95	0.6	0.54	0.68
IndicBERT	Denoised + MFCC	VGG16 + ELA	RF	74.99	0.8	0.71	0.89
IndicBERT	Denoised + MFCC	VGG16 + ELA	SVM	67.18	0.61	0.55	0.61
IndicBERT	Denoised + MFCC	VGG16 + ELA	DT	70.8	0.68	0.73	0.63
IndicBERT	Denoised + MFCC	VGG16 + ELA	LR	74.5	0.78	0.85	0.69
IndicBERT	Denoised + MFCC	VGG16 + ELA	XGBOOST	75.04	0.79	0.81	0.71
IndicBERT	Denoised + MFCC	VGG16 + ELA	ADABOOST	76.47	0.68	0.77	0.76
IndicBERT	Denoised + MFCC	ResNet + ELA	RF	70.75	0.67	0.71	0.72
IndicBERT	Denoised + MFCC	ResNet + ELA	SVM	71.93	0.65	0.6	0.71
IndicBERT	Denoised + MFCC	ResNet + ELA	DT	69.04	0.65	0.66	0.64
IndicBERT	Denoised + MFCC	ResNet + ELA	LR	70.57	0.7	0.62	0.64
IndicBERT	Denoised + MFCC	ResNet + ELA	XGBOOST	74.85	0.78	0.87	0.75
IndicBERT	Denoised + MFCC	ResNet + ELA	ADABOOST	70.57	0.66	0.64	0.6
MuRIL	HuBERT	ViT	RF	56.27	0.61	0.6	0.59
MuRIL	HuBERT	ViT	SVM	61.75	0.71	0.78	0.77
MuRIL	HuBERT	ViT	DT	53.3	0.5	0.55	0.56
MuRIL	HuBERT	ViT	LR	61.94	0.64	0.59	0.54
MuRIL	HuBERT	ViT	XGBOOST	55.82	0.54	0.58	0.59
MuRIL	HuBERT	ViT	ADABOOST	53.06	0.53	0.54	0.62
MuRIL	HuBERT	VGG16	RF	60.99	0.69	0.74	0.78
MuRIL	HuBERT	VGG16	SVM	61.99	0.59	0.65	0.52
MuRIL	HuBERT	VGG16	DT	58.84	0.53	0.6	0.61
MuRIL	HuBERT	VGG16	LR	53.1	0.6	0.5	0.57
MuRIL	HuBERT	VGG16	XGBOOST	55.63	0.53	0.58	0.52
MuRIL	HuBERT	VGG16	ADABOOST	58.93	0.54	0.6	0.56
MuRIL	HuBERT	ResNet	RF	61.22	0.61	0.59	0.62
MuRIL	HuBERT	ResNet	SVM	59.56	0.53	0.6	0.57
MuRIL	HuBERT	ResNet	DT	54.98	0.52	0.62	0.56
MuRIL	HuBERT	ResNet	LR	57.51	0.57	0.61	0.59
MuRIL	HuBERT	ResNet	XGBOOST	56.81	0.58	0.53	0.55
MuRIL	HuBERT	ResNet	ADABOOST	58.92	0.66	0.75	0.76
MuRIL	HuBERT	ViT + ELA	RF	57.3	0.56	0.61	0.57
MuRIL	HuBERT	ViT + ELA	SVM	59.81	0.6	0.59	0.68
MuRIL	HuBERT	ViT + ELA	DT	56.58	0.57	0.51	0.67
MuRIL	HuBERT	ViT + ELA	LR	55.77	0.65	0.63	0.63
MuRIL	HuBERT	ViT + ELA	XGBOOST	61.18	0.52	0.51	0.58
MuRIL	HuBERT	ViT + ELA	ADABOOST	56.44	0.53	0.55	0.53
MuRIL	HuBERT	VGG16 + ELA	RF	53.43	0.57	0.51	0.64
MuRIL	HuBERT	VGG16 + ELA	SVM	57.59	0.54	0.51	0.57
MuRIL	HuBERT	VGG16 + ELA	DT	59.87	0.59	0.68	0.5
MuRIL	HuBERT	VGG16 + ELA	LR	55.28	0.62	0.59	0.63

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
MuRIL	HuBERT	VGG16 + ELA	XGBOOST	53.51	0.59	0.57	0.53
MuRIL	HuBERT	VGG16 + ELA	ADABOOST	60.6	0.55	0.58	0.54
MuRIL	HuBERT	ResNet + ELA	RF	53.46	0.63	0.62	0.59
MuRIL	HuBERT	ResNet + ELA	SVM	58.29	0.56	0.65	0.5
MuRIL	HuBERT	ResNet + ELA	DT	59.01	0.68	0.6	0.7
MuRIL	HuBERT	ResNet + ELA	LR	58.18	0.58	0.52	0.53
MuRIL	HuBERT	ResNet + ELA	XGBOOST	59.54	0.52	0.61	0.53
MuRIL	HuBERT	ResNet + ELA	ADABOOST	57.97	0.52	0.62	0.6
MuRIL	MFCC	ViT	RF	59.26	0.56	0.57	0.51
MuRIL	MFCC	ViT	SVM	60.65	0.64	0.56	0.65
MuRIL	MFCC	ViT	DT	58.98	0.59	0.66	0.57
MuRIL	MFCC	ViT	LR	53.04	0.6	0.59	0.67
MuRIL	MFCC	ViT	XGBOOST	60.46	0.63	0.62	0.7
MuRIL	MFCC	ViT	ADABOOST	54.76	0.61	0.69	0.51
MuRIL	MFCC	VGG16	RF	60.67	0.63	0.67	0.65
MuRIL	MFCC	VGG16	SVM	53.66	0.61	0.57	0.6
MuRIL	MFCC	VGG16	DT	55.07	0.56	0.58	0.52
MuRIL	MFCC	VGG16	LR	59.07	0.59	0.68	0.62
MuRIL	MFCC	VGG16	XGBOOST	56.52	0.51	0.59	0.55
MuRIL	MFCC	VGG16	ADABOOST	56.63	0.52	0.55	0.58
MuRIL	MFCC	ResNet	RF	61.31	0.55	0.58	0.54
MuRIL	MFCC	ResNet	SVM	53.23	0.63	0.67	0.59
MuRIL	MFCC	ResNet	DT	60.73	0.66	0.59	0.72
MuRIL	MFCC	ResNet	LR	57.09	0.53	0.54	0.58
MuRIL	MFCC	ResNet	XGBOOST	56.46	0.58	0.65	0.57
MuRIL	MFCC	ResNet	ADABOOST	60.23	0.6	0.6	0.53
MuRIL	MFCC	ViT + ELA	RF	60.34	0.67	0.65	0.76
MuRIL	MFCC	ViT + ELA	SVM	56.05	0.63	0.63	0.55
MuRIL	MFCC	ViT + ELA	DT	59.26	0.52	0.53	0.62
MuRIL	MFCC	ViT + ELA	LR	56.72	0.51	0.59	0.53
MuRIL	MFCC	ViT + ELA	XGBOOST	56.35	0.62	0.68	0.67
MuRIL	MFCC	ViT + ELA	ADABOOST	53.75	0.51	0.59	0.6
MuRIL	MFCC	VGG16 + ELA	RF	60.58	0.6	0.67	0.6
MuRIL	MFCC	VGG16 + ELA	SVM	54.96	0.64	0.72	0.65
MuRIL	MFCC	VGG16 + ELA	DT	60.12	0.64	0.74	0.65
MuRIL	MFCC	VGG16 + ELA	LR	59.68	0.52	0.52	0.59
MuRIL	MFCC	VGG16 + ELA	XGBOOST	55.31	0.51	0.57	0.51
MuRIL	MFCC	VGG16 + ELA	ADABOOST	57.3	0.51	0.54	0.6
MuRIL	MFCC	ResNet + ELA	RF	60.95	0.68	0.77	0.74
MuRIL	MFCC	ResNet + ELA	SVM	55.63	0.61	0.59	0.69
MuRIL	MFCC	ResNet + ELA	DT	54.54	0.51	0.55	0.55
MuRIL	MFCC	ResNet + ELA	LR	59.46	0.69	0.71	0.69
MuRIL	MFCC	ResNet + ELA	XGBOOST	57.04	0.63	0.66	0.54
MuRIL	MFCC	ResNet + ELA	ADABOOST	56.35	0.59	0.67	0.65
MuRIL	Denoised + HuBERT	ViT	RF	53.26	0.59	0.67	0.53
MuRIL	Denoised + HuBERT	ViT	SVM	53.48	0.6	0.51	0.51
MuRIL	Denoised + HuBERT	ViT	DT	60.18	0.62	0.64	0.55
MuRIL	Denoised + HuBERT	ViT	LR	55.01	0.64	0.73	0.59
MuRIL	Denoised + HuBERT	ViT	XGBOOST	59.57	0.55	0.56	0.51
MuRIL	Denoised + HuBERT	ViT	ADABOOST	59.61	0.57	0.53	0.64
MuRIL	Denoised + HuBERT	VGG16	RF	53.17	0.57	0.64	0.59
MuRIL	Denoised + HuBERT	VGG16	SVM	56.78	0.6	0.51	0.55
MuRIL	Denoised + HuBERT	VGG16	DT	57.14	0.51	0.61	0.55

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
MuRIL	Denoised + HuBERT	VGG16	LR	57.7	0.55	0.57	0.54
MuRIL	Denoised + HuBERT	VGG16	XGBOOST	57.34	0.66	0.59	0.63
MuRIL	Denoised + HuBERT	VGG16	ADABOOST	53.73	0.55	0.61	0.57
MuRIL	Denoised + HuBERT	ResNet	RF	56.37	0.65	0.68	0.61
MuRIL	Denoised + HuBERT	ResNet	SVM	60.66	0.7	0.77	0.75
MuRIL	Denoised + HuBERT	ResNet	DT	53.15	0.55	0.53	0.53
MuRIL	Denoised + HuBERT	ResNet	LR	57.81	0.62	0.63	0.69
MuRIL	Denoised + HuBERT	ResNet	XGBOOST	55.14	0.58	0.52	0.65
MuRIL	Denoised + HuBERT	ResNet	ADABOOST	60.19	0.62	0.59	0.52
MuRIL	Denoised + HuBERT	ViT + ELA	RF	61.12	0.62	0.58	0.56
MuRIL	Denoised + HuBERT	ViT + ELA	SVM	61.87	0.6	0.57	0.54
MuRIL	Denoised + HuBERT	ViT + ELA	DT	59.46	0.62	0.72	0.57
MuRIL	Denoised + HuBERT	ViT + ELA	LR	58.55	0.63	0.6	0.62
MuRIL	Denoised + HuBERT	ViT + ELA	XGBOOST	57.88	0.66	0.75	0.57
MuRIL	Denoised + HuBERT	ViT + ELA	ADABOOST	56.42	0.61	0.7	0.71
MuRIL	Denoised + HuBERT	VGG16 + ELA	RF	59.66	0.69	0.69	0.77
MuRIL	Denoised + HuBERT	VGG16 + ELA	SVM	54.61	0.54	0.61	0.53
MuRIL	Denoised + HuBERT	VGG16 + ELA	DT	60.97	0.58	0.68	0.56
MuRIL	Denoised + HuBERT	VGG16 + ELA	LR	61.28	0.56	0.53	0.65
MuRIL	Denoised + HuBERT	VGG16 + ELA	XGBOOST	57.19	0.62	0.53	0.61
MuRIL	Denoised + HuBERT	VGG16 + ELA	ADABOOST	54.91	0.63	0.62	0.56
MuRIL	Denoised + HuBERT	ResNet + ELA	RF	58.36	0.65	0.56	0.58
MuRIL	Denoised + HuBERT	ResNet + ELA	SVM	61.36	0.56	0.64	0.64
MuRIL	Denoised + HuBERT	ResNet + ELA	DT	59.38	0.6	0.54	0.51
MuRIL	Denoised + HuBERT	ResNet + ELA	LR	53.85	0.57	0.57	0.66
MuRIL	Denoised + HuBERT	ResNet + ELA	XGBOOST	58.23	0.5	0.55	0.57
MuRIL	Denoised + HuBERT	ResNet + ELA	ADABOOST	53.63	0.62	0.6	0.56
MuRIL	Denoised + MFCC	ViT	RF	53.04	0.6	0.65	0.53
MuRIL	Denoised + MFCC	ViT	SVM	57.09	0.58	0.55	0.67
MuRIL	Denoised + MFCC	ViT	DT	55.71	0.62	0.68	0.7
MuRIL	Denoised + MFCC	ViT	LR	61.29	0.56	0.52	0.6
MuRIL	Denoised + MFCC	ViT	XGBOOST	58.88	0.51	0.53	0.5
MuRIL	Denoised + MFCC	ViT	ADABOOST	59.91	0.6	0.56	0.57
MuRIL	Denoised + MFCC	VGG16	RF	61.99	0.6	0.64	0.58
MuRIL	Denoised + MFCC	VGG16	SVM	57.03	0.62	0.66	0.71
MuRIL	Denoised + MFCC	VGG16	DT	58.13	0.68	0.62	0.78
MuRIL	Denoised + MFCC	VGG16	LR	56.46	0.52	0.57	0.5
MuRIL	Denoised + MFCC	VGG16	XGBOOST	55.17	0.5	0.54	0.55
MuRIL	Denoised + MFCC	VGG16	ADABOOST	57.13	0.67	0.59	0.6
MuRIL	Denoised + MFCC	ResNet	RF	54.16	0.51	0.57	0.56
MuRIL	Denoised + MFCC	ResNet	SVM	60.55	0.55	0.64	0.58
MuRIL	Denoised + MFCC	ResNet	DT	58.39	0.63	0.7	0.71
MuRIL	Denoised + MFCC	ResNet	LR	55.75	0.65	0.71	0.55
MuRIL	Denoised + MFCC	ResNet	XGBOOST	57.96	0.67	0.69	0.68
MuRIL	Denoised + MFCC	ResNet	ADABOOST	56.18	0.6	0.55	0.58
MuRIL	Denoised + MFCC	ViT + ELA	RF	55.15	0.5	0.59	0.51
MuRIL	Denoised + MFCC	ViT + ELA	SVM	57.52	0.53	0.56	0.57
MuRIL	Denoised + MFCC	ViT + ELA	DT	58.48	0.57	0.6	0.56
MuRIL	Denoised + MFCC	ViT + ELA	LR	56.06	0.64	0.67	0.7
MuRIL	Denoised + MFCC	ViT + ELA	XGBOOST	55.1	0.58	0.56	0.66
MuRIL	Denoised + MFCC	ViT + ELA	ADABOOST	60.34	0.54	0.58	0.57
MuRIL	Denoised + MFCC	VGG16 + ELA	RF	60.29	0.52	0.57	0.51
MuRIL	Denoised + MFCC	VGG16 + ELA	SVM	58.18	0.58	0.63	0.68

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
MuRIL	Denoised + MFCC	VGG16 + ELA	DT	58.56	0.61	0.55	0.69
MuRIL	Denoised + MFCC	VGG16 + ELA	LR	58.99	0.66	0.68	0.7
MuRIL	Denoised + MFCC	VGG16 + ELA	XGBOOST	61.1	0.53	0.53	0.53
MuRIL	Denoised + MFCC	VGG16 + ELA	ADABOOST	60.23	0.67	0.76	0.63
MuRIL	Denoised + MFCC	ResNet + ELA	RF	58.43	0.58	0.65	0.52
MuRIL	Denoised + MFCC	ResNet + ELA	SVM	55.9	0.63	0.62	0.58
MuRIL	Denoised + MFCC	ResNet + ELA	DT	56.81	0.6	0.56	0.66
MuRIL	Denoised + MFCC	ResNet + ELA	LR	58.83	0.67	0.7	0.66
MuRIL	Denoised + MFCC	ResNet + ELA	XGBOOST	59.36	0.62	0.59	0.65
MuRIL	Denoised + MFCC	ResNet + ELA	ADABOOST	61.21	0.56	0.59	0.64
LaBSE	HuBERT	ViT	RF	49.74	0.54	0.52	0.56
LaBSE	HuBERT	ViT	SVM	51.31	0.52	0.52	0.56
LaBSE	HuBERT	ViT	DT	50.74	0.6	0.53	0.61
LaBSE	HuBERT	ViT	LR	52.31	0.57	0.56	0.58
LaBSE	HuBERT	ViT	XGBOOST	51.34	0.5	0.55	0.58
LaBSE	HuBERT	ViT	ADABOOST	50.22	0.59	0.62	0.59
LaBSE	HuBERT	VGG16	RF	58.99	0.59	0.51	0.68
LaBSE	HuBERT	VGG16	SVM	57.6	0.66	0.67	0.61
LaBSE	HuBERT	VGG16	DT	54.85	0.58	0.6	0.56
LaBSE	HuBERT	VGG16	LR	56.63	0.65	0.64	0.64
LaBSE	HuBERT	VGG16	XGBOOST	55.18	0.61	0.53	0.57
LaBSE	HuBERT	VGG16	ADABOOST	53.49	0.62	0.58	0.52
LaBSE	HuBERT	ResNet	RF	57.71	0.51	0.56	0.6
LaBSE	HuBERT	ResNet	SVM	58.32	0.61	0.56	0.54
LaBSE	HuBERT	ResNet	DT	52.57	0.55	0.51	0.56
LaBSE	HuBERT	ResNet	LR	54.3	0.52	0.52	0.54
LaBSE	HuBERT	ResNet	XGBOOST	49.28	0.55	0.6	0.63
LaBSE	HuBERT	ResNet	ADABOOST	57.82	0.55	0.51	0.62
LaBSE	HuBERT	ViT + ELA	RF	49.02	0.5	0.6	0.52
LaBSE	HuBERT	ViT + ELA	SVM	55.52	0.56	0.54	0.59
LaBSE	HuBERT	ViT + ELA	DT	50.01	0.6	0.57	0.63
LaBSE	HuBERT	ViT + ELA	LR	49.93	0.57	0.66	0.66
LaBSE	HuBERT	ViT + ELA	XGBOOST	57.41	0.6	0.6	0.56
LaBSE	HuBERT	ViT + ELA	ADABOOST	53.03	0.57	0.55	0.61
LaBSE	HuBERT	VGG16 + ELA	RF	57.42	0.64	0.66	0.57
LaBSE	HuBERT	VGG16 + ELA	SVM	58.95	0.69	0.72	0.72
LaBSE	HuBERT	VGG16 + ELA	DT	56.4	0.63	0.64	0.55
LaBSE	HuBERT	VGG16 + ELA	LR	48.51	0.57	0.61	0.55
LaBSE	HuBERT	VGG16 + ELA	XGBOOST	49.29	0.51	0.61	0.59
LaBSE	HuBERT	VGG16 + ELA	ADABOOST	58.36	0.67	0.62	0.63
LaBSE	HuBERT	ResNet + ELA	RF	50	0.6	0.5	0.64
LaBSE	HuBERT	ResNet + ELA	SVM	48.16	0.55	0.62	0.58
LaBSE	HuBERT	ResNet + ELA	DT	54.38	0.58	0.52	0.62
LaBSE	HuBERT	ResNet + ELA	LR	57.01	0.62	0.56	0.54
LaBSE	HuBERT	ResNet + ELA	XGBOOST	54.65	0.62	0.63	0.57
LaBSE	HuBERT	ResNet + ELA	ADABOOST	51.31	0.56	0.58	0.59
LaBSE	MFCC	ViT	RF	48.17	0.52	0.55	0.58
LaBSE	MFCC	ViT	SVM	51.94	0.53	0.55	0.55
LaBSE	MFCC	ViT	DT	52.58	0.56	0.56	0.51
LaBSE	MFCC	ViT	LR	51.55	0.52	0.53	0.6
LaBSE	MFCC	ViT	XGBOOST	56.08	0.66	0.71	0.57
LaBSE	MFCC	ViT	ADABOOST	57.49	0.67	0.57	0.71
LaBSE	MFCC	VGG16	RF	55.79	0.65	0.57	0.65

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
LaBSE	MFCC	VGG16	SVM	55.16	0.64	0.63	0.65
LaBSE	MFCC	VGG16	DT	55.33	0.57	0.51	0.66
LaBSE	MFCC	VGG16	LR	52.07	0.51	0.61	0.6
LaBSE	MFCC	VGG16	XGBOOST	55.17	0.59	0.66	0.56
LaBSE	MFCC	VGG16	ADABOOST	49.72	0.59	0.52	0.52
LaBSE	MFCC	ResNet	RF	55.74	0.58	0.65	0.61
LaBSE	MFCC	ResNet	SVM	57.48	0.6	0.59	0.66
LaBSE	MFCC	ResNet	DT	53.76	0.55	0.61	0.54
LaBSE	MFCC	ResNet	LR	50.28	0.54	0.58	0.55
LaBSE	MFCC	ResNet	XGBOOST	52.59	0.55	0.64	0.53
LaBSE	MFCC	ResNet	ADABOOST	51.52	0.55	0.6	0.58
LaBSE	MFCC	ViT + ELA	RF	58.81	0.59	0.6	0.65
LaBSE	MFCC	ViT + ELA	SVM	52.32	0.61	0.56	0.59
LaBSE	MFCC	ViT + ELA	DT	55.71	0.55	0.52	0.54
LaBSE	MFCC	ViT + ELA	LR	58.46	0.53	0.58	0.59
LaBSE	MFCC	ViT + ELA	XGBOOST	51.36	0.52	0.52	0.54
LaBSE	MFCC	ViT + ELA	ADABOOST	56.8	0.5	0.54	0.55
LaBSE	MFCC	VGG16 + ELA	RF	54.44	0.61	0.56	0.65
LaBSE	MFCC	VGG16 + ELA	SVM	57.45	0.59	0.63	0.54
LaBSE	MFCC	VGG16 + ELA	DT	48.83	0.54	0.56	0.6
LaBSE	MFCC	VGG16 + ELA	LR	55.58	0.61	0.6	0.6
LaBSE	MFCC	VGG16 + ELA	XGBOOST	51.29	0.51	0.59	0.53
LaBSE	MFCC	VGG16 + ELA	ADABOOST	56.65	0.56	0.61	0.64
LaBSE	MFCC	ResNet + ELA	RF	48.68	0.51	0.58	0.56
LaBSE	MFCC	ResNet + ELA	SVM	56.98	0.5	0.52	0.54
LaBSE	MFCC	ResNet + ELA	DT	58.52	0.66	0.64	0.61
LaBSE	MFCC	ResNet + ELA	LR	55.83	0.57	0.65	0.64
LaBSE	MFCC	ResNet + ELA	XGBOOST	51.5	0.58	0.57	0.68
LaBSE	MFCC	ResNet + ELA	ADABOOST	57.05	0.5	0.57	0.51
LaBSE	Denoised + HuBERT	ViT	RF	55.69	0.56	0.62	0.53
LaBSE	Denoised + HuBERT	ViT	SVM	58.81	0.63	0.58	0.62
LaBSE	Denoised + HuBERT	ViT	DT	55	0.62	0.61	0.57
LaBSE	Denoised + HuBERT	ViT	LR	48.46	0.54	0.58	0.52
LaBSE	Denoised + HuBERT	ViT	XGBOOST	58.71	0.69	0.75	0.78
LaBSE	Denoised + HuBERT	ViT	ADABOOST	58.98	0.55	0.53	0.64
LaBSE	Denoised + HuBERT	VGG16	RF	49.02	0.5	0.5	0.51
LaBSE	Denoised + HuBERT	VGG16	SVM	57.37	0.57	0.63	0.54
LaBSE	Denoised + HuBERT	VGG16	DT	48.04	0.51	0.6	0.61
LaBSE	Denoised + HuBERT	VGG16	LR	56.99	0.6	0.61	0.63
LaBSE	Denoised + HuBERT	VGG16	XGBOOST	48.79	0.52	0.53	0.54
LaBSE	Denoised + HuBERT	VGG16	ADABOOST	51.97	0.59	0.58	0.67
LaBSE	Denoised + HuBERT	ResNet	RF	57.23	0.53	0.52	0.58
LaBSE	Denoised + HuBERT	ResNet	SVM	55.76	0.57	0.67	0.67
LaBSE	Denoised + HuBERT	ResNet	DT	53.23	0.59	0.61	0.66
LaBSE	Denoised + HuBERT	ResNet	LR	48.52	0.52	0.55	0.52
LaBSE	Denoised + HuBERT	ResNet	XGBOOST	57.38	0.56	0.65	0.6
LaBSE	Denoised + HuBERT	ResNet	ADABOOST	50.54	0.53	0.52	0.56
LaBSE	Denoised + HuBERT	ViT + ELA	RF	50.08	0.58	0.62	0.53
LaBSE	Denoised + HuBERT	ViT + ELA	SVM	54.62	0.5	0.54	0.58
LaBSE	Denoised + HuBERT	ViT + ELA	DT	50.71	0.54	0.51	0.63
LaBSE	Denoised + HuBERT	ViT + ELA	LR	53.9	0.52	0.59	0.52
LaBSE	Denoised + HuBERT	ViT + ELA	XGBOOST	54.1	0.55	0.59	0.54
LaBSE	Denoised + HuBERT	ViT + ELA	ADABOOST	48.55	0.55	0.58	0.62

TABLE 7. *(Continued.)* All 96 experiment results.

Text	Speech	Image	Classifier	Accuracy	F1 Score	Precision	Recall
LaBSE	Denoised + HuBERT	VGG16 + ELA	RF	57.57	0.55	0.58	0.61
LaBSE	Denoised + HuBERT	VGG16 + ELA	SVM	52.61	0.62	0.68	0.57
LaBSE	Denoised + HuBERT	VGG16 + ELA	DT	51.7	0.51	0.58	0.61
LaBSE	Denoised + HuBERT	VGG16 + ELA	LR	51.06	0.52	0.59	0.54
LaBSE	Denoised + HuBERT	VGG16 + ELA	XGBOOST	56.46	0.66	0.7	0.59
LaBSE	Denoised + HuBERT	VGG16 + ELA	ADABOOST	57.62	0.64	0.56	0.56
LaBSE	Denoised + HuBERT	ResNet + ELA	RF	49.12	0.51	0.53	0.53
LaBSE	Denoised + HuBERT	ResNet + ELA	SVM	51.04	0.6	0.59	0.6
LaBSE	Denoised + HuBERT	ResNet + ELA	DT	57.78	0.53	0.62	0.62
LaBSE	Denoised + HuBERT	ResNet + ELA	LR	54.58	0.56	0.54	0.55
LaBSE	Denoised + HuBERT	ResNet + ELA	XGBOOST	48.2	0.52	0.54	0.62
LaBSE	Denoised + HuBERT	ResNet + ELA	ADABOOST	54.22	0.51	0.61	0.53
LaBSE	Denoised + MFCC	ViT	RF	57.03	0.5	0.59	0.54
LaBSE	Denoised + MFCC	ViT	SVM	52.64	0.55	0.63	0.58
LaBSE	Denoised + MFCC	ViT	DT	49.08	0.52	0.55	0.61
LaBSE	Denoised + MFCC	ViT	LR	52.68	0.57	0.62	0.51
LaBSE	Denoised + MFCC	ViT	XGBOOST	55.43	0.51	0.56	0.58
LaBSE	Denoised + MFCC	ViT	ADABOOST	50.73	0.56	0.5	0.63
LaBSE	Denoised + MFCC	VGG16	RF	51.41	0.57	0.66	0.57
LaBSE	Denoised + MFCC	VGG16	SVM	51.43	0.6	0.66	0.65
LaBSE	Denoised + MFCC	VGG16	DT	58.84	0.62	0.72	0.63
LaBSE	Denoised + MFCC	VGG16	LR	54.84	0.56	0.59	0.54
LaBSE	Denoised + MFCC	VGG16	XGBOOST	49.82	0.59	0.56	0.58
LaBSE	Denoised + MFCC	VGG16	ADABOOST	55.31	0.56	0.61	0.52
LaBSE	Denoised + MFCC	ResNet	RF	54.23	0.54	0.5	0.63
LaBSE	Denoised + MFCC	ResNet	SVM	49.8	0.59	0.63	0.61
LaBSE	Denoised + MFCC	ResNet	DT	58.37	0.61	0.58	0.55
LaBSE	Denoised + MFCC	ResNet	LR	58.49	0.63	0.54	0.6
LaBSE	Denoised + MFCC	ResNet	XGBOOST	56.02	0.58	0.51	0.65
LaBSE	Denoised + MFCC	ResNet	ADABOOST	52.39	0.54	0.51	0.6
LaBSE	Denoised + MFCC	ViT + ELA	RF	49.7	0.51	0.52	0.54
LaBSE	Denoised + MFCC	ViT + ELA	SVM	55.04	0.62	0.59	0.64
LaBSE	Denoised + MFCC	ViT + ELA	DT	53.39	0.52	0.56	0.57
LaBSE	Denoised + MFCC	ViT + ELA	LR	49.29	0.5	0.59	0.6
LaBSE	Denoised + MFCC	ViT + ELA	XGBOOST	56.08	0.58	0.66	0.59
LaBSE	Denoised + MFCC	ViT + ELA	ADABOOST	57.65	0.6	0.53	0.51
LaBSE	Denoised + MFCC	VGG16 + ELA	RF	50.31	0.58	0.51	0.6
LaBSE	Denoised + MFCC	VGG16 + ELA	SVM	51.51	0.61	0.6	0.59
LaBSE	Denoised + MFCC	VGG16 + ELA	DT	50.84	0.56	0.56	0.55
LaBSE	Denoised + MFCC	VGG16 + ELA	LR	58.01	0.67	0.71	0.66
LaBSE	Denoised + MFCC	VGG16 + ELA	XGBOOST	56.75	0.53	0.54	0.57
LaBSE	Denoised + MFCC	VGG16 + ELA	ADABOOST	53.81	0.54	0.54	0.61
LaBSE	Denoised + MFCC	ResNet + ELA	RF	55.94	0.52	0.59	0.53
LaBSE	Denoised + MFCC	ResNet + ELA	SVM	49.85	0.55	0.53	0.58
LaBSE	Denoised + MFCC	ResNet + ELA	DT	50.47	0.55	0.53	0.62
LaBSE	Denoised + MFCC	ResNet + ELA	LR	51.07	0.55	0.64	0.51
LaBSE	Denoised + MFCC	ResNet + ELA	XGBOOST	52.34	0.52	0.55	0.58
LaBSE	Denoised + MFCC	ResNet + ELA	ADABOOST	58.04	0.63	0.73	0.69

dataset from fact-checking platforms that operate within this specific context. Hence, the dataset effectively encapsulates the political nuances that determine how people perceive and trust news in Tamil Nadu.

Finally, the content from these fact-checking websites often includes direct quotes, screenshots, and analyses of viral social media posts. This content exposes our models to the linguistic features unique to Tamil social media discourse, such as informal language, transliteration practices (Tanglish), and slang. By training on this authentic content, our multimodal models implicitly learn to identify patterns linked to the linguistic and stylistic traits of misinformation spread on Tamil social media. While our current work primarily advances technical aspects of multimodal fusion for detection, the inherent characteristics of our dataset establish a strong foundation. This foundation allows for future analyses to delve deeper into these socio-cultural and linguistic dimensions. By leveraging data from active Tamil fact-checking initiatives, our models learn directly from real-world, context-specific misinformation challenges in the Tamil language.

B. ANALYSIS OF EXPERIMENTAL FINDINGS AND BROADER IMPLICATIONS

Previous research predominantly focused on text-image combinations or binary classification schemes, creating a substantial gap in the literature. For instance, prior Tamil fake news detection systems typically achieved F1 scores below 75% using only textual features. Our integration of speech as a third modality, incorporating 884 minutes of audio data, represents a methodological breakthrough for low-resource language context. Furthermore, the four-label classification approach (true, fake, misleading, and altered) aligns more closely with real-world misinformation patterns compared to binary classifications. This enhanced taxonomy acknowledges that social media content often exists on a broad spectrum between completely true and entirely false.

The t-SNE visualizations across combinations of all models consistently show distinct clustering patterns for each modality. This confirms that the proposed multimodal architecture successfully captures complementary information rather than redundant features. This supports our claim that speech features provide critical contextual cues that text or images alone cannot capture in Tamil content.

Initial experiments revealed a significant class imbalance challenge, with predictions disproportionately labeled as Fake. The confusion matrix Figure. 7 highlighted this bias stemmed from the uneven dataset distribution—a common issue in fact-checking datasets where debunked content is overrepresented. Furthermore, implementing cost-sensitive learning successfully rebalanced class weights, substantially reducing false FAKE predictions while improving precision and recall for minority classes (true, misleading, and altered), as evidenced in Figure. 8. We observed a 15% improvement in F1 score, which indicates that traditional machine learning models struggle without explicitly addressing class imbalance in linguistically complex datasets.

The top-performing embedding combination, IndicBERT + HuBERT + ViT + ELA, excels due to IndicBERT's strong semantic capture from its pretraining on a vast Tamil corpus, enabling better handling of code-mixed social media text. HuBERT complements this by providing robust self-supervised speech representations that detect prosody and acoustic subtleties in Tamil audio. ViT captures global image dependencies for complex visuals, while ELA enhances forgery detection, together boosting recall in subtle misinformation classes. This feature fusion reduces modality conflicts, yielding a 85% F1 score that sets a baseline benchmark in our low-resource multimodal Tamil fake news classification tasks.

Despite these advances, several limitations merit acknowledgement. Our study has limitations regarding the modalities included and the dataset composition. Our use of synthetic speech to train our model limits its ability to generalize to real-world conditions. We used clean, uniform TTS audio, which does not contain speaker diversity, emotional cues, and background noise that characterize authentic audio. Therefore, the model's performance on natural speech data remains untested. And, the exclusion of the video modality means our analysis could not address multimodal misinformation involving deepfakes, which are prevalent on social media. Secondly, the dataset is imbalanced, containing significantly more 'Fake' than 'True' samples. This reflects the focus of our data sources various fact-checking organizations on debunking viral misinformation rather than verifying truthful content. While our work establishes initial benchmarks, comprehensive fake news detection systems should incorporate more balanced class distributions. Additionally, the overfitting curves across all models show some degree of validation performance decline after epoch 10, suggesting potential for further optimization through regularization techniques beyond hyperparameter tuning.

Three priorities emerge for follow-up research. First, developing explainable AI frameworks that highlight why specific content is flagged (e.g., identifying manipulated speech segments or mismatched image-text contexts) would enhance user trust; the distinct modality clusters observed in our t-SNE visualizations provide a foundation for such interpretation mechanisms. Second, sarcasm-aware models leveraging Tamil cultural references and meme databases could address a significant gap in current systems that frequently misclassify satirical content. Third, real-time fact-checking browser extensions with multilingual support would enable the practical application of our findings, particularly for Tamil speakers worldwide.

We acknowledge that automated fake-news detectors can be misused for censorship or to disproportionately silence minority and opposition voices. During an election, a government rolls out the model to automatically flag "fake news" and uses a low confidence threshold. As a result, satirical memes and legitimate critiques of ruling-party policies are mislabeled as "fake" and removed without human review, disproportionately silencing opposition voices and journalists. Our system is a research prototype meant to assist fact-checkers not to make standalone moderation decisions and should not be used for fully automated fact-checking. To reduce risks, we require human review for impactful actions, use calibrated confidence thresholds with an "uncertain/abstain" option, and conduct regular bias audits.

This research establishes the first comprehensive multimodal baseline (text-speech-image) for Tamil fake news detection, achieving an 85% F1 Score on a four-label taxonomy. By solving class imbalance through cost-sensitive learning and releasing a public dataset, we create a foundation for scalable misinformation combat tools tailored to Tamil-language content. The results underscore that regional language models require both multimodal inputs and cultural context awareness a framework readily adaptable to other low-resource languages such as Malayalam, Telugu, and Kannada facing similar challenges in combating digital misinformation.

VI. CONCLUSION

In this paper, we have studied the problem of multimodal fake news classification and introduced a robust framework for classifying fake news in the low-resource language Tamil. This framework integrates "IndicBERT + HuBERT + ViT+ELA", which emerges as the benchmark baseline model for a multimodal embedding with an F1 score of 85%. Further, we developed the first multimodal dataset, Tamil-facts, which consists of data from various modalities such as text, images and speech. We conducted a comprehensive analysis of misinformation in Tamil. The proposed approach also addressed the problem of class imbalance through cost-sensitive learning. The experimental results demonstrated that it improved the precision and recall of minority classes. This study marks a critical step toward combating misinformation in India's diverse digital landscape. At the same time, the publicly available dataset empowers researchers to create more models tailored to regional contexts. Future work will focus on incorporating natural speech samples of misinformation, analysing video modality, predominance of fake-labeled samples in the dataset and to develop a categorically annotated version of our dataset, which will enable a more fine-grained investigation into the model's effectiveness on specific domains such as politics, healthcare, and entertainment. We are withholding the dataset and codebase to preserve the integrity of our long-term project. Committed to reproducible science, we will release the whole dataset and associated code once the broader research concludes.

APPENDIX

EXTENDED RESULTS TABLE

See Table 7.

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